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**"INVESTIGATION ON DRY ROOT ROT OF CHICKPEA
INCITED BY *Rhizoctonia bataticola* (Taub) Butler."**

BY
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B.Sc. (Agri.)

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COLLEGE OF AGRICULTURE, BADNAPUR
VASANTRAO NAIK MARATHWADA KRISHI VIDYAPEETH,
PARBHANI 431 402 (M.S.), INDIA
2015**

**“INVESTIGATION ON DRY ROOT ROT OF
CHICKPEA INCITED BY *Rhizoctonia bataticola*
(Taub) Butler.”**

DISSERTATION

*Submitted to the
Vasantrao Naik Marathwada Krishi Vidyapeeth,
Parbhani*

*In partial fulfillment of the requirements
for the Degree of*



**MASTER OF SCIENCE
(Agriculture)**

IN

PLANT PATHOLOGY

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2015**

CANDIDATES DECLARATION

I hereby declare that the dissertation

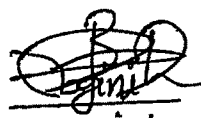
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Place : Badnapur



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Date: 7 / 8 / 2015

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CERTIFICATE – I

This is to certify that the dissertation entitled “**Investigation on Dry root rot of Chickpea incited by *Rhizoctonia bataticola* (Taub) Butler.**” Submitted by **Ms. Date Yogini Bhausaheb Reg.No. 20MB/2013A** to the **College of Agriculture, Badnapur** in partial fulfillment of the requirements for the degree of **MASTER OF SCIENCE (Agriculture)** in the subject of **PLANT PATHOLOGY** is record of original and bonafide research work carried out by her under my guidance and supervision. It is of sufficiently high standard to warrant its presentation for the award of the said degree.

I also certify that the dissertation or part thereof has not been previously submitted by her for degree of any university.

Place: Badnapur
Date: 7 / 8 / 2015


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Research Guide and
Chairman of Advisory committee

CERTIFICATE – II

This is to certify that the dissertation entitled “**Investigation on Dry root rot of Chickpea incited by *Rhizoctonia bataticola* (Taub) Butler.**” Submitted by **Ms.Date Yogini Bhausaheb Reg.No. 20MB/2013A** to the **College of Agriculture, Badnapur** in partial fulfillment of the requirements for the degree of **MASTER OF SCIENCE (Agriculture)** in the subject of **PLANT PATHOLOGY** has been approved by the student’s advisory committee after viva-voce examination in collaboration with the external examiner.


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Place: Badnapur

Date:

(Date Y. B.)

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ABBREVIATIONS

%	-	Per cent
@	-	At the rate of
C.D.	-	Critical difference
cm	-	Centimeter (s)
DI	-	Disease Incidence
e.g.	-	Exempli Gratis (for Example)
<i>et al.</i>	-	And others
etc.	-	Etcetera
Fig.	-	Figure (s)
ha	-	Hectares (s)
hrs.	-	Hours
i.e.	-	That is
kg	-	Kilogram (s)
m	-	Meter (s)
Max.	-	Maximum
Min.	-	Minimum
mm	-	Millimeter
No.	-	Number (s)
°C	-	Degree Celsius
PDA	-	Potato Dextrose Agar
PDI	-	Per cent Disease Intensity
S.E.	-	Standard error
sp.	-	Species
T	-	Treatment
viz.,	-	Videlicet (namely)

APPENDIX-I

1. PREPARATION OF CULTURE MEDIA

1.1 Potato Dextrose Agar (1 lit.)

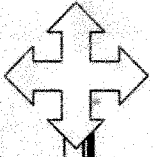
Potato : 200 g
Dextrose : 20 g
Agar Agar : 20g
Distilled water: 1000ml

1.2 Sand maize flour medium (1 lit.)

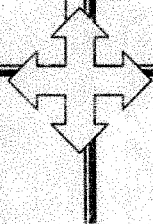
Maize flour : 10 g
Sand : 90 g
Distilled water: 1000mS

2.0 CONCENTRATION USED IN FORMULATION

1	Per cent	= 10,000 ppm (parts per million)	
		= 10 gms/ liter	
		= 10 gms/ kg	
0.1	Per cent	= 1000 ppm	= 1000 mg/liiter
0.01	Per cent	=100 ppm	= 100 mg/liiter
0.001	Per cent	=10 ppm	= 10 mg/liiter
0.0001	Per cent	= 1 ppm	= 1 mg/liiter



INTRODUCTION



CHAPTER I

INTRODUCTION

The cultivated chickpea (*Cicer arietinum* L.) was one of the first grain legume to be domesticated in the old world. Chickpea is most probably originated in area of present day south eastern Turkey and adjoin Syria.

Historically the name chickpea dates back to French 'Chiche' and to Latin *Cicer* from which Roman Cognomen Cicero was taken but the Oxford English Dictionary lists a 1548 citation indicate, "*Cicer* may be named in English Cich, or ciche pease, after the Frenche tonge." The original name "Chick-pea" that is printed in English in 1388 was taken directly from French as cited by the dictionary in the Mid 18th century (Zohary and Maria, 2000).

The origin of chickpea is still controversial as postulated by several botanists in different geographical origins. De Candolle (1883), identified that the region between the South of the Caucasus and in the North of Persia are the place of possible origin of chickpea as this was also supported by van der Maesen, 1972. Vavilov (1926) supported the ideas that the South west Asia and the Mediterranean region as the primary centers of origin with Ethiopia as the secondary center. The medicinal value of the chickpea is worth mentioning here also the leaves and seeds of the chickpea due to the presence of glandular secretions are commonly used as medicine. This plant holds a good repute in 'Ayurvedic' and 'Unani' system of medicine and according to Ayurvedic method of treatment, chickpea leaves are sour, astringent to bowels, and improve taste and appetite. Moreover, the leaves are used to cure chronic bronchitis and the seeds are considered as antibilious, used as tonic, stimulant and aphrodisiac acid is also supposed to lower the blood cholesterol level (Duke, 1981). Due to its aphrodisiac properties, it is referred as Vajibhakshya in Sanskrit (Oudhia, 2003). Chickpea has also the property to act as hypocholesteremic agent; germinating chickpea is believed to reduce the blood

cholesterol level. Exudation of leaves locally called Amb contain malic acid and oxalic acid which posses medicinal value for bronchitis, constipation, diarrhea, digestive disorder, warts and blood purification. Germinated seed are recommended to cure scurvy disease.

In India Chickpea is primarily grown as *rabi* (post rainy) season crop on residual soil moisture. It improves physical, chemical and biological properties of soil by fixing atmospheric nitrogen symbiotically. Deep roots of pulse crop also open up the soil by increasing soil aeration and fit well in various cropping ecosystem and hence have got unique position in reinfed agriculture.

Chickpea is grown in many tropical, sub-tropical and temperate regions of the world. It is the best and cheapest source of protein in human diet. It is the rich source of protein 21.1 per cent carbohydrate 61.5 per cent and fat 4.5 per cent. Chickpea is mainly used for human consumption as well as animal feed. It is consumed as whole seed, dal, fried, boiled, salted or more generally cooked fresh green leaves are used as vegetable. The grains also used as vegetable chole. Gram flour is mixed with wheat flour to improve the protein content of wheat flour and it is used for making missi roti. The flour of dehusked gram is called besan which is widely used in making pakodas, kadhi and several snack foods.

Chickpea is grown throughout the world. In india it is grown on large scale in Punjab, Haryana, Uttar Pradesh, Madhya Pradesh, Rajasthan, Andhra Pradesh, Karnataka and Maharashtra. It is mostly cultivated under rainfed condition in a variety of soil, varying in residual moisture. In India, area under chickpea was 9.51 million ha. with production 8.88 metric ton and productivity 929 kg/ha during 2013-14. In Maharashtra chickpea is grown under *rabi* season and it occupies area of 11345 thousand ha with production 8676 thousand ton and productivity 765 kg/ha during 2013-14. In Marathawada

region, total area under chickpea during 2013-14 was 3788 thousand ha with production 2522 thousand ton and productivity 599 kg/ha

On the basis of food characteristics demand system, the projection of the year 2005, 2010 and 2020 are 20.0, 23.3 and 27.0 mt respectively (Anonymous,2002). However, chickpea productivity is low due to susceptibility of the crop to different biotic and abiotic stresses. Abiotic stress is basically due to insufficient moisture. Regarding biotic stresses, diseases, insect pests, nematodes and parasitic weeds account major losses. The chickpea crop is attacked by 172 pathogen such as 67 fungi, 22 viruses, 3 bacteria, 80 nematodes and mycoplasma from all over the world (Nene *et al.*,1996). Among all only a few of them have the potential to devastate the crops. Some of the serious disease in order of their importance are wilt (*fusarium oxysporum f. sp. ciceri*), Dry root rot (*Rhizoctonia bataticola*) Ascochyta blight(*Ascothya rabici*) and Collar rot (*Sclerotium rolfsie*). Among all dry root rot caused by *Rhizoctonia bataticola* (Taub) Butler is one of the major soil born disease of chickpea .The pathogen survive in soil in infected host debris for several year. Secondary spread is through farm implements, irrigation water and rain splashes.

Chickpea dry root rot causing pathogen *Rhizoctonia bataticola* belongs to group of ubiquitous and diverse plant pathogens that occur widely in India as a root pathogen on different crops and also isolated from different varieties of same host species. Natural population of this pathogen is diverse in terms of pathogenic variability and can be characterized into pathotypes or races by means of biological typing.

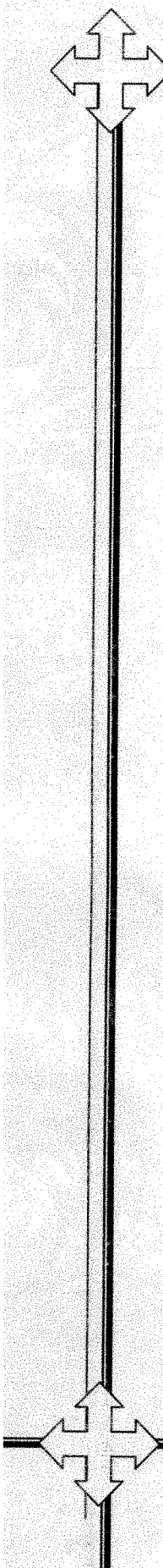
Being soil borne in nature it is very difficult to control this pathogen. Use of resistant cultivars is more effective strategy for sustainable chickpea cultivation. Despite of several resistant varieties available, yield losses due to root rot are more. Ahmed and Mohammed (1986) reported losses

due to dry root rot in gram to extent of 70.8 percent at full podding and 48.9 per cent at preharvest stage. It indicates that use of resistant varieties is not sole control measure against dry root rot of chickpea. Therefore, two or more control measure must be tried in combination with resistant variety for sustainable production.

Considering the behavior of pathogen, losses incurred in the farmers field and limitations of individual control measure the present investigation was planned with following objectives.

OBJECTIVES:

- 1) To record the incidence of dry root rot of chickpea in field of Marathawada region.
- 2) To isolate and prove the pathogenicity of *Rhizoctonia bataticola*
- 3) To test the efficacy of different fungicides against mycelial growth of *Rhizoctonia bataticola* (poisoned food technique)
- 4) To test the efficacy of different botanicals against mycelial growth of *Rhizoctonia bataticola* (Poisoned food technique)
- 5) To test the efficacy of fungal bioagents against *Rhizoctonia bataticola* by dual culture method.
- 6) To test the efficacy of seed treatment with fungicides, botanicals, and bioagents on germination and per cent incidence of dry root rot (Pot culture)
- 7) To screen different varieties and germplasm against *Rhizoctonia bataticola* in dry root rot.



REVIEW OF LITERATURE

CHAPTER-II

REVIEW OF LITERATURE

Chickpea (*Cicer arintinum* L.) is one of the important pulse crop grown in India and it is largely grown as rainfed as well as irrigated crop in *rabi* season. Production of chickpea is affected by many soil borne pathogens. An attempt has been made in this chapter to compile all the relevant literature pertaining to dry root rot of chickpea and its management under following sub-headings.

1. Occurrence
2. Yield losses
3. Survey on dry root rot
4. Collection, Isolation, Pathogenicity
5. Symptomatology
6. Disease management
7. Screening of Chickpea varieties, Germplasm line and Genotype

2.1 Occurrence

Agrawal and Sarbhoy (1976) reported that that widespread occurrence of *Rhizoctonia bataticola* (*M. phaseolina*), causing charcoal rot of Soybean and observed that the pathogen infects the crop in almost all stages of crop growth.

Smith and Carvil (1977) reported that *Rhizoctonia bataticola* (*M. phaseolina*) causes seedling blight, root rot of more than 500 crop and non crop species.

Saksena (1979) reported that *Rhizoctonia bataticola* (*M. phaseolina*) was the wide spread and destructive plant pathogen. The pathogen had been considered of great economic importance because, the pathogen was reported to cause root rot, collar rot, stalk rot, leaf spot and blight.

Vishwadhar and Sarbhoy (1989) reported that *Rhizoctonia bataticola* (Taub) Butler (Pycnidial stage, *M. phaseolina*) caused seedling rot, root rot, ashey stem blight and charcoal rot in the crops like soybean, mung bean, sunflower, urdbean etc. in India.

Srivastava *et al.* (2002) conducted survey in the Vindhyan plateaux zone (Bhopal, Raisen, Rajgarh, Sagar, Sehore and Vidisha) of Madhya Pradesh, India, during 1998/99 to evaluate the prevalence of wilt and dry root diseases in chickpea. Some 50 plants showing wilt symptoms were collected from 23 locations. The greatest incidence of wilt was recorded for Sehore (25per cent), followed by Rajgarh, Bhopal and Vidisha (20 per cent). Five fungi (*Fusarium oxysporum*, *Rhizoctonia bataticola* [*Macrophomina phaseolina*], *Trichoderma* sp., *Aspergillus* sp. and *Curvularia* sp.) were associated with chickpea samples. *F. oxysporum* was the most prevalent (56 per cent) in wilted plants, followed by *R. bataticola* (17per cent) and *Trichoderma* sp. (10 per cent). The highest incidence of *F. oxysporum* was detected in samples from Bhopal and Sehore (67per cent), followed by Raisen (60 per cent), Vidisha and Sagar (50 per cent), and Rajgarh (40 per cent). The incidence of *R. bataticola*, which also caused withering and drying of plants, ranged from 8 to 20 per cent. The occurrence of *Trichoderma* was higher in samples with lower incidence of *F. oxysporum* and vice versa.

Singh and Sirohi (2003) observed that dry root rot caused by *R. bataticola* was a serious disease in non irrigated chickpea areas. The incidence of disease was more in Una and Simour district of Himachal Pradesh.

Gurha and Trivedi (2008) reported that *F. oxy.f.sp.ciceri* and *R. bataticola* caused wilting and dry root rot diseases respectively in all districts of Himachal Pradesh.

Rhiaz *et al.* (2011) conducted survey in Jammu and Kashmir, Dry root rot disease caused by *Rhizoctonia bataticola* was observed on variety Radhey in Jammu Kashmir.

2.2 Yield losses

Quantitative and qualitative yield losses incurred by root rot disease in chickpea, groundnut, black gram, green gram, cluster bean, sunflower; sesammum etc. were reported earlier by several workers.

Nene (1985) reported that chickpea root rot, collar rot and wilt were the most serious disease which cause severe yield losses of 30-60 per cent under favorable condition.

Quesier and Mohammad (1986) reported that yield losses due to *Rhizoctonia* root rot of chickpea in Bihar state and observed that the disease caused about 50 to 71 per cent yield losses in chickpea.

Sen (2000) reported that *Macrophomina phaseolina* affected a variety of field crops and vegetable crops besides a wide range of urdbean plant species and caused charcoal rot, which accounted for crop losses to the tune of 10-20 per cent.

Vishwadhar and Chaudhary (2001) reported that chickpea was susceptible to several fungal diseases amongst them, Dry root rot caused by *Macrophomina phaseolina* was one of the major production constraints that caused 10-20 per cent annual losses.

2.3 Survey on dry root rot of chickpea

Maheswari *et al.* (1983) conducted on survey of various pea growing areas in North India and found losses up to 13.9 -95 per cent due to dry root rot.

Sharma *et al.*, (1983) conducted on survey of wilt and root rot incidence of *Cicer arietinum* in northern region of Madhya Pradesh. Survey was conducted on Chickpea growing areas of 30 villages of northern region of Madhya Pradesh. The average of wilt incidence ranged from 14.90 to 37.99 per cent and root rot from 3.58 -20.36 per cent.

Pande and Singh (1990) conducted survey in Allahabad region, India and observed Chickpea to be infected by *F.oxysporum* and *R. bataticola*.

Srivastava *et al.* (2002) conducted survey in Vidhyan plateau zone of Madhya Pradesh, during 1998-99 to evaluate the prevalence of wilt and dry root rot diseases in Chickpea. The incidence of *R. bataticola*, which also caused withering and drying of plants, ranged from 8 to 20 per cent.

Prajapati *et al.* (2003) conducted survey on incidence of Chickpea dry root rot in Uttar Pradesh and Uttaranchal and reported that the incidence of dry root rot of chickpea caused by (*Macrophomina phaseolina*) at 20 locations in Uttar Pradesh and Uttaranchal.

Prajapati *et al.* (2004) conducted survey on incidence of Chickpea dry root rot in Uttar Pradesh. Survey was conducted at 20 locations at different agro-

climatic regions of Uttar Pradesh. The disease was wide spread in occurrence in UP with 19.0 to 42.0 per cent incidence.

Kohire *et al.* (2006) carried out a survey in 2004-2006. They reported that wilt, dry root rot and stunt were associated with infected chickpea in Marathawada region of Maharashtra. Disease incidence varied from 1.6 to 16.5 per cent. Maximum average incidence was observed in Parbhani 18.5 per cent.

Rana *et al.* (2009) conducted survey on pea diseases and identification of fungi associated with wilt and root rot complex in Himachal Pradesh. The disease were prevalent in all pea growing areas of state during 2006-2008. The incidence of wilt and root rot was 20 per cent in temperate wet, 15 per cent in sub temperate, 12 per cent in temperate dry and 11 per cent in subtropical regions.

2.4 Collection, Isolation and Pathogenicity

Bhate (1983) isolated *Rhizoctonia bataticola* from roots of green gram and proved its pathogenicity.

Sobti and Bansal (1988) isolated root, leaf and seed isolates of *R. bataticola* of groundnut and reported the variation in cultural characters on different media.

Bansal *et al.* (1990) reported cultural variability among six isolates of *R. bataticola* isolated from different crops.

Vishwadhar and Sarbhoy (1991) studied on the typical isolates of *R. bataticola* (*M. phaseolina*) reported that, *R. bataticola* (*M. phaseolina*) was isolated from pods and seeds of few diseased Soybean plant.

Devi and Singh (1998) studied on virulence of *Macrophomina phaseolina* isolates from black gram and green gram and reported that, fifty six

Macrophomina phaseolina isolates were obtained from roots, stems, pods and seeds of Black gram and Green gram from 11 different localities of North, South, and Central India. All isolates were categorized into different groups.

Patel and Anahosur (2000) found association frequency of soil born fungi with wilted Chickpea plants. Isolation made at 10 days interval from 10th day of sowing up to 90th day revealed preferential association of *Fusarium spp.*, *M. phaseolina* (Tassi) Goid and *Sclerotium rolfsii* sacc.

Katariya *et al.* (2007) isolated nine isolates of *R. bataticola* and tested pathogenicity of these isolates both in sterilized and unsterilized soil in Chickpea variety C-325.

Thippeswamy *et al.* (2007) studied on pathogenicity and management of brown lesion and leaf spot in Okra caused by *M. phaseolina*. *Alternaria alternate* reported that the pathogenicity test and management was carried out at the department experimental plot. *M. phaseolina*, showed 10-20 per cent in two month seedling and 1-10 per cent in two month seedling and 10-20 per cent in three months old plant showing brown lesion spots.

2.5 Symptomatology

Nene *et al.* (1978) found continuous black discoloration of pith and xylem vessels of roots and basal shoots as one of the main characteristic symptoms of in gram infected by *R. bataticola*.

Casas and Diaz (1985) described symptoms caused by *Macrophomina phaseolina* as dry collar and root rot. Affected plants developed dark brown to black roots and dried up without showing any flaccidity.

Haware *et al.* (1990) reported that the disease development was influenced by dry soil condition especially at flowering stage. The plant suddenly dries in the field. The leaves and stem of affected plants usually become straw colored. The tap root is dark and quite brittle in dry soil. The dark black sclerotia can be observed on and within host tissue.

Beniwal *et al.* (1992) reported sudden drying of the whole plant was the most conspicuous symptom of this disease. The tap root of infected plant become dark showed signs of rotting, and is devoid of most of the lateral and finer roots. The dead root was brittle and its tip was easily broken.

2.6 Disease management strategies.

2.6.1 Evaluation of fungicide

Shukla and Singh (1973) studied the efficacy of fungicidal seed treatment on *Macrophomina* root rot of Sesame and revealed that plots sown with Captan treated seed gave highest yield.

Chauhan *et al.* (1988) observed that seed treatment or soil application of Carbendazim followed by Carboxin gave good disease control against pre and post emergence seedling mortality of soil borne fungal pathogen complex of cotton.

Gupta and Cheema (1990) reported that Thiram and Captan increased the seedling stand and reduced the disease incidence in case of *M. phaseolina* of sesame.

Taya *et al.* (1990) reported that fungicide reduced dry root rot incidence as compared to control. Carbendazim brought down disease incidence to 28.80 per cent followed by mixture of Carbendazim and Thiram when applied as

seed treatment out of which eight fungicides applied as soil drench before sowing Carbendazim alone reduced disease incidence from 86.60 to 35.27 per cent followed by mixture of Carbendazim and Thiram (35.50 per cent).

Saxena and Saxena (1993) reported that infection caused by *F. oxy.f.sp.lintil*, *S.rolfsii* and *R.bataticola* (*M.phaseolina*) in lentil was reduced by seed treatment with Thiram+Carbendazim (2:1, 2.5 g/kg seed).

Rathore (2001) reported that seed treatment with Bavistin 50 WP recorded minimum percent disease intensity and maximum seed yield followed Captan 75 SD against dry root rot of cluster bean.

Prajapati *et al.*(2002) reported that Carbendazim 25 DS and Carbendazim 50WP + thiram 80 WP completely inhibited the growth of *R.bataticola* and were found to be most effective.

Sharma and Gupta (2004) studied efficacy of the fungicide in the control of Chickpea dry root rot caused by *Rhizoctonia bataticola* and reported that seed treatment with Carbendazim (1 gm /kg seed) showed the minimum incidence of disease (17-21 per cent) and increased grain.

Vijay –Mohan *et al.*(2006) studied the fungicidal management of dry root rot disease and yield of chickpea and reported that Carbendazim (0.2 per cent) and Etaconazole (0.1 per cent) used as seed treatment , soil drenching and combined effect recorded lowest disease incidence of 15.6 and 18.2 per cent and highest grain yield of 192 and 18.9 q/ha respectively.

Kaushal and Sood (2008) studied management of root rot in chickpea and reported that seed treatment with Thiram and Bavistin @ 2.0gm/ kg seed gave good control of root rot of chickpea under field condition.

2.6.2 Evaluation of Bioagents

Parkhia and Vishnav (1986) studied biological control of *Macrophomina phaseolina* and reported that the introduction of *Trichoderma harzianum* reduced the disease to an appreciable level in comparison to check. In seed treatment, 18 per cent plants were infected with *R. bataticola*, in soil drenching 28 per cent and wheat husk barn culture 14 per cent as compared to 70 per cent in check.

Selverajan and Jeyaraan (1996) screening eight fungi and bacteria *in vitro* to select suitable antagonists for Chickpea roots rot caused by *Fusarium solani* and *M. phaseolina*. *T. harzianum*, *T. hamatum*, *Lectisaria arvalis*, *B. subtilis* and *P. fluorescence* formed inhibition zone against both fungi.

Singh *et al.* (1998) studied on application of antagonists in relation to dry root rot and biochemical status in Chickpea plants and reported that all the antagonists used as seed treatment soil inoculation and soil inoculation with seed treatment were significantly superior in reducing incidence of disease. All antagonists *T. harzianum*, *T. viride*, *P. fluorescence*, *A. flavus* and *B. subtilis* except *A. flavus* were not effective for controlling dry root rot of Chickpea.

Malathi and Doraisamy (2003) reported that *Trichoderma spp.* tested, significantly inhibited the pathogen growth and the inhibition ranged from 32 to 67 per cent. The maximum inhibition was shown by *T. harzinum-5* followed by *T. koningii*.

Jyotsana *et al.* (2008) assessed strains of *T. harzianum* for their effect on chickpea growth and control of charcoal rot caused by *M. phaseolina* in green house assay. *T. harzianum* strain 25-92 significantly increased the fresh and dry weight by 50.63 per cent and 24.42 per cent respectively, whereas strain 29-92

increased the fresh weight of Chickpea cvs Radhey and Vishwas by 12.30 per cent. Significant reduction in charcoal rot was observed in pots amended with *T.harzianum* at all concentrations.

Kaushal (2008) reported that *T.harzianum* was effective in inhibiting the mycelial growth of *R.bataticola* the causal organism of Chickpea dry root rot.

Khan and Gangopadhy (2008) studied efficacy of *Pseudomonas fluorescens* in controlling root rot of Chickpea and reported that the root rot incidence was significantly reduced due to seed treatment with PFBC-25 and PFBC-26. Although the disease incidence reduced with an increase in dose from 4 to 8 gm/kg seed.

Pan (2009) reported the antagonistic potential of *Trichoderma* isolate through production of volatile substance against *Macrophomina phaseolina*.

Chaudhary *et al.* (2010) reported that minimum incidence of root rot was observed in *Trichoderma harzianum* treated plants followed by pots treated with *T.viride*. Incidence of root rot was higher in *Bacillus firmus* significantly effective in reducing the dry root rot incidence in Mung bean plants *T. viride* and *T.harzianum* and *A. versicolor* were observed effective in the suppression of dry root rot incidence.

2.6.3 Evaluation of Botanicals

Shivpuri *et al.* (1997) reported that ethanol extracts of *Allium sativum*, *Azadirachta indica*, *Calotropis procera*, *Datura stramonium*, *Tagetes erecta*, *Vinca rosea* and *Withania somnifera* possessed fungitoxic properties against *Fusarium oxysporum*, *Rhizoctonia bataticola* and *Sclerotinia sclerotiorum* at 500 and 1000 ppm concentration.

Saju *et al.* (1998) reported antifungal properties of essential oils of Turmeric against *Fusarium oxysporum*, *Rhizoctonia bataticola*, *Colletotrichum gloesporioides*, *Pestotia palmarum* and *Aspergillus* spp. All concentrations of oil from 1-5 per cent inhibited the growth of *Colletotrichum gloesporioides* and *Pestotia palmarum* by 100 per cent, *Rhizoctonia bataticola* by 79 per cent and *Aspergillus* spp. by 53.2 per cent.

Datar (1999) studied the effect of garlic (*Allium sativum*) extract on charcoal rot in Sorghum. Garlic was found most inhibitory to *R. bataticola*.

Mandhare and Suryawanshi (2009) studied *in vitro* and revealed that the extract of *Azadirachta indica* (80 per cent) produced maximum inhibition against root rot of chickpea.

Tandel *et al.* (2010) tried phyto-extracts of eleven plant species on *M. phaseolina* of green gram and revealed that the onion bulb extract produced maximum inhibition (98.14 per cent) followed by extract of Acacia, Ginger, Neem, Garlic and Karanj.

2.7 Screening

Agarwal and Sarbhoy (1976) evaluated of Soybean germplasm for resistance against *Macrophomina phaseolina*, reported that sixty-four lines were found to moderately resistant and four lines viz., EC-1077, EC-25166, EC-39069 and EC-39797 resistant under artificially inoculated field as well as glass house condition.

Krishnamohan *et al.* (1981) tested 20 chickpea forms under field conditions and artificially in the green house, using a 1-6 score (1-highly resistant), BG205 and BG206, although moderately susceptible in the green house, were, highly resistant under field conditions against *R. bataticola*

Singh *et al.* (1982) reported that 3 chickpea cultivars did not to be related to their resistance to *R. bataticola* carbohydrate content was higher in the susceptible cultivar than in the 2 resistant ones.

Singh *et al.* (1989) studied sesamum germplasm for resistant against *Macrophomina* root rot reported that out of 312 cultivar lines were tested in which 30 lines were found resistant; 158 moderately resistant; 124 showed susceptible reactions. The resistant lines were AK-77, ES-4, IS-175, KIS-430, N66-3169, SI-1740, SI-1153, SI-1875.

Reddy *et al.* (1990) reported additional sources of resistance to wilt and root rot in chickpea from ICRISAT, Hyderabad. Out of which 1883 accessions evaluated 177 accessions showed less than 20 percent mortality in the first year of screening and of these 40 accessions showed less than 10 per cent mortality in second year of screening.

Lodha and Solanki (1993) studied on inheritance of resistance to dry root rot in Cluster bean and reported that the susceptibility in the genotypes varied from 22.3 to 46.6 per cent. However, least (26.1 per cent) pre-emergences mortality was recorded in the HFG-156 Followed by IC-922913 (33.9 per cent) while Kutch-8 recorded lowest (22.3 per cent), Post emergence mortality followed by RGC-471 and highly susceptible varieties HG-75 (40.5 per cent) and AG-111 (46.6 per cent) were identified for their use in resistant breeding programme.

Zote *et al.* (1993) screened Chickpea germplasm lines against wilt and dry root rot pathogen and reported BCP-28, BCP-29 and BCP-36 lines were resistant to *fusarium* wilt and lines BCP-15, BCP-23, BCP-29, and BCP-30 were resistance to both wilt and dry root rot.

Gupta (1995) identified chickpea genotypes with dual resistance against wilt and root rot. He found that mortality in susceptible check JG -62

ranged from 80-100 per cent. Genotype ICCV-89402, ICCV-90201, ICCV-90254, BG-209, H88-2, GF88426 and RSG-180 suffered less than 10 per cent incidence of both the diseases.

Sajeena *et al.* (2004) studied on screening of Black gram varieties against root rot caused by *Macrophomina phaseolina* reported that among the seventeen genotypes of Black gram tested BG ADT 5 and BGKI recorded the maximum germination and yield and root rot disease incidence was at a minimum. IC14520 and IC14691 registered the minimum germination and yield and showed maximum disease incidence. None of the genotypes were found to be completely resistant to root rot disease.

Dinkaran *et al.* (2005) studied on screening of Sesame genotypes against root rot pathogen *Macrophomina phaseolina* reported that among the 1300 Sesame genotype screened only three accessions viz., RM-7, ORM-14 and ORM-17, showed resistant reaction.

Mishra *et al.* (2005) have tested 470 germplasm lines and found KG-86 KWR-4, KWR-108 and KWR-277 as a resistant genotype against dry root rot.

Jayalakshmi *et al.* (2008) screened different genotypes for their resistance to dry root rot in pot culture experiment with susceptible check JG-62 during *rabi* season of 2004-2005 and 2005-2006. Out of 12 genotypes tested against root rot disease, 4 were found resistant, 2 were moderately susceptible, 3 were susceptible and 3 were highly susceptible to dry root rot.

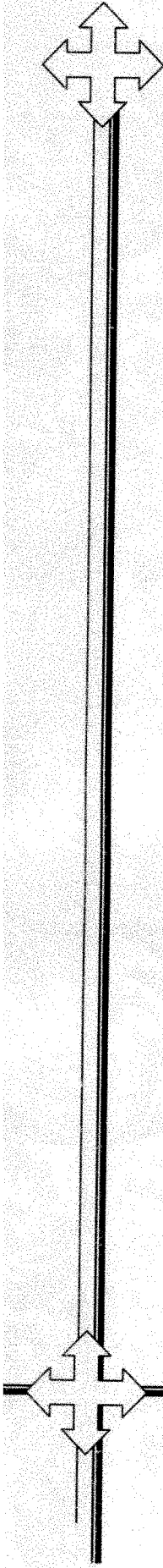
Aghakhani *et al.* (2009) twenty – three isolates of *R. bataticola* causing dry root rot of chickpea (*Cicer arietinum*) collected from 10 different major chickpea growing states of India were highly variable in their morphological and cultural characters as well as pathogenicity /virulence. The virulence analysis of the isolates on a set of chickpea cultivars namely ICC12441,

ICC1224, ICC12450, Pusa 362, BGD112, Pusa1103, Pusa212, Pusa1088 and under blotter paper as well as sick soil grouped them into 6 pathotypes. The pathotype groups were related to agro ecological region of the country. The most virulent isolate (RBI from Bangalore, Karnataka) was fast growing and produced largest sclerotia. A set of cultivars was proposed for the first time for differentiating the pathotypes of *R. bataticola* causing dry root rot of chickpea.

Chaturvedi and Dua (2009) have reported 25 resistant cultivars including KPG-59, Radhey and K-50 against dry root rot.

Ved Ratan *et al.* (2010) reported that the variation in date of sowing as an effective and economic strategy against dry root rot caused by (*R. bataticola*) and wilt (*Fusarium oxysporum f.sp. ciceris*) disease of chickpea

Riyaz *et al.* (2012) reported management of chickpea dry root rot through biocontrol agents and resistant germplasm lines in Jammu Kashmir.



MATERIAL AND METHODS

CHAPTER –III

MATERIAL AND METHOD

3.1 Experimental materials

The various kinds of materials viz., seeds, fertilizers, fungicides, chemicals, glassware's, culture media and other miscellaneous items required for conducting present studies will be obtained from the Department of Plant Pathology, College of Agriculture, Badnapur and Agricultural Research Station, Badnapur.

3.1.1 Disease samples

Plant showing typical symptoms of chickpea dry root rot collected from farmer's fields belonging to Marathawada region.

3.1.2 Culture media

Common laboratory media i.e. Potato Dextrose Agar (PDA) used for isolation of pathogen from infected plant sample.

Composition of media

Potato Dextrose Agar Medium

Peeled potato	200.00g
Dextrose	20.00g
Agar	20.00g
Distilled water	1000 ml

3.1.3 Seeds

Genetically pure seeds of chickpea obtained from Agricultural Research centre, Badnapur.

3.1.4 Chemicals

The chemicals ingredients, reagents etc. used for the preparation of media were obtained from Agricultural Research centre Badnapur.

3.1.5 Biological agents

The isolates of *T. viride*, *T. harzianum*, *T. hamatum*, *T. longibrachitum*, *T. koningii*, *Aspergillus niger*, *Aspergillus flavus*, *Pseudomonas fluorescens* were obtained from the Department of Plant Pathology, College of Agriculture, Parbhani..

3.1.6 Fungicides

The following fungicides were used in pot culture studies under glass house condition against *Rhizoctonia bataticola*.

1. Carbendazim

- a) Chemical name : 2 (methoxy-carbamoyl)-benzimidazole
- b) Active ingredient : 50% wettable powder
- c) Concentration used : 0.2%
- d) Manufacturer (s): BASF India Ltd., Mumbai.

2. Captan

- a) Chemical name : 2 (methoxy-carbamoyl)-benzimidazole
- b) Active ingredient : 50% wettable powder
- c) Concentration used : 0.3%
- d) Manufacturer (s): Rallis India Ltd., Mumbai.

3. Thiram

- a) Chemical name : Tetramethyl thiram disulphate
- b) Active ingredient : 75% wettable powder
- c) Concentration used : 0.3%
- d) Manufacturer (s): Bharat Pulverising mills Ltd., Mumbai.

4. Mancozeb

- a) Chemical name : 1,2- Ethazneiybis (carbamodithio)(2-) Maganese zink salt
- b) Active ingredient : 75% wettable powder
- c) Concentration used : 0.15%

d) Manufacturer (s): Hindustan Pulverising mills Ltd., Jammu.

5. SAAF

a) Chemical name: Carbendazim (12%)+Mancozeb(63%)

b) Active ingredient: 63% wettable powder

c) Concentration used: 0.30 %

6. Copper oxychloride

a) Chemical name: Copper chloride oxide hydrate

b) Active ingredient: 50% wettable powder

c) Concentration used: 0.15%

d) Manufacturer (s): Chaitanya Agro chemicals, Ahmedabad

7. Zineb

a) Chemical name : Zinc ethylenebis dithiocarbamate

b) Active ingredient: 75% wettable powder

c) Concentration used: 0.15%

d) Manufacturer (s): Bhaskar Agrochemical limited information, Hyderabad

3.1.7 Botanicals/Plant extracts

Mehandi, Ginger, Parthenium, Neem, Garlic, Turmeric, Bougainvillea
Ghaneri samples were collected from field of Agriculture Research
Station, Badnapur

3.1.8 Field and laboratory facilities

All field experiment will be planned and conducted on the
research farm of the Department of Plant Pathology, College of Agriculture,
Badnapur and Agricultural Research Station, Badnapur. *In vitro* studies will be
conducted in the laboratory and screen house of this department.

3.1.9 Glasswares

Different types of corning and Borosil brand glassware were used
in the present investigations. The common glasswares used were petriplates,

test tubes, conical flask, measuring cylinder, glass rod, spirit jar, beakers and puppets.

3.1.10 Equipments

The laboratory equipments used such as Autoclave, B.O.D. incubator, Refrigerator, Shaker, Research microscope, and analytical scientific balance etc.

3.1.11 Miscellaneous materials

Inoculating needle, forceps, blotter paper, spirit lamp, mercuric chloride, cork borer, plastic bucket, distilled water, earthen pot, paper bags, polythene bags, labels, scales, soil isolation chamber, laminar airflow cabinet, were used during the course of present investigations which are available in Plant pathology section , College of Agriculture, Badnapur. And Agriculture research station, Badnapur.

3.2 Methodology

3.2.1 Survey

A field survey of chickpea dry root rot disease was conducted in eight district of Marathawada region during the month of December and February, 2014-15 to record occurrence and distribution of chickpea dry root rot disease. On an average 10 farmer's field of chickpea in each district were visited and the percent dry root rot disease incidence was recorded by counting total chickpea plant $1 \times 1 \text{m}^2$ area and total dry root rot infected plants in that area. Chickpea plants showing typical symptoms were collected in separate paper bags and brought to the laboratory for investigations. The intensity of disease was noted by counting at least 400 plants in each field. Observation on dry root rot intensity was recorded as, the symptoms expressed as straw colored plant like dry grass at pod formation , black rotted

roots , shredding of bark and root broken easily, minute dark black sclerotial bodies on root surface and in the pith region.(PLATE- I)

The per cent disease incidence was calculated by using 0-9 point disease rating scale. On an average ten villages of each district were surveyed for dry root rot severity, chickpea cultivar and disease incidences recorded are presented in below table.

List of Chickpea growers in Marathwada region

Sr. No	Name of farmer	District	Village	Stage of crop	Variety
1	Prabhakar Pawar	Aurangabad	Shivoor	Podding	Vijay
2	Ganesh Pawar		Devgaon	Podding	Vijay
3	Appasaheb Jadhav		Shivoor	Flowering	Vijay
4	Sainath Jadhav		Shivoor	Podding	BDN 797
5	Devidas Kite		Shivoor	Flowering	Vijay
6	Pandurang Pawar		Shivoor	Podding	Vijay
7	Uttam Madake		Shivoor	Podding	Digvijay
8	Sanjay Tupe		Kolani	Podding	Virat
9	Rajendra Jadhav		Ganori	Podding	Virat
10	Anil Mohite		Kannad	Podding	Virat
11	Bhauasaheb Jadhav		Gangapur	Podding	Virat
12	Maruti Totare	Nanded	pala	Flowering	local
13	Khandu Totare		Pala	Flowering	Local
14	Rama Totare		pala	Flowering	Vijay
15	Eknath Totare		pala	Podding	Vijay
16	Dattatraya Gaikwad		Ganjgaon	Podding	Vijay
17	Shankar Gaikwad		Ganjgaon	Podding	Vijay
18	Subhash Gaikwad		Ganjgaon	Podding	Virat
19	Rajiv Gaikwad		Ganjgaon	Podding	Local
20	Rajanna Surkutlavar		Sagaroli	Flowering	Local
21	Rajendrasingh Chauhan		Sagaroli	Podding	Local
22	Vitthal Surkutlavar		Sagaroli	Podding	Local
23	Vankat surkotalavar		Sagaroli	Podding	Local
24	Ashokrao Rodage	Parbhani	Ravalgaon	Podding	BDN 797
25	Pratap Rodage		Ravalgaon	Podding	BDN 797

26	Shivaji Rodage		Ravalgaon	Podding	Virat
27	Suresh Rodage		Ravalgaon	Podding	Virat
28	Bhaskar Rodage		Ravalgaon	Flowering	Vijay
29	Sachin Kadam		Purna	Flowering	Vijay
30	Shriram Khandare		Jintur	Flowering	Local
31	Ganeshrao Deshmukh		Parbhani	Flowering	Local
32	Bapusaheb Deshmukh		Parbhani	Flowering	Digvijay
33	Manik Game	Jalana	Parbhani	Podding	Digvijay
34	Ramesh Bhopale		Rajur	Podding	Digvijay
35	Rameshvar Kankhar		Deulgaon	Podding	Virat
36	Baban Jadhav		Deulgaon	Podding	Virat
37	Ganesh Bhopale		Khasgaon	Podding	Vijay
38	Ambadas Kanhere		Gevrai bazaar	Podding	Vijay
39	Ramesh Lahane		Gevrai bazaar	Podding	BDN 797
40	Prabhakar Joshi		Gevrai bazaar	Podding	Local
41	Appasaheb Kanhere		Gevrai bazaar	Podding	Local
42	Raosaheb Fatale		kandari	Podding	Vijay
43	Sandu Ugale		Deulgaon	Podding	Vijay
44	Sadashiv Fatale		Kandari	Podding	Virat
45	Ram Fatale		Kandari	Podding	Virat
46	Rajiv Fatale		kandari	Podding	Vijay
47	Sanjivani Pandule	Beed	Pimpri ghumari	Flowering	Local
48	Sagar Pandule		Pimpri	Podding	Virat
49	Asurba Mutkule		Ashti	Podding	Local
50	Vijay Pokale		Ashti	Podding	Digvijay
51	Ajinath Pandule		Pimpri	Podding	Vijay
52	Ashabai Pandule		Pimpri	Podding	Vijay
53	Rahul Jagatap		Ashti	Podding	Vijay
54	Sunil Mutkule		Pimpri	Podding	BDN797
55	Vikas Pandule		Pimpri	Podding	Virat
56	Ambadas Jagtap		Pimpri	Podding	Virat
57	Shivajirao Mane	Latur	nilanga	flowering	Vijay
58	Vasantarao patil		Arvi	Flowering	Vijay
59	Ambadas Nade		Murud	flowering	Vijay
60	Vivek Shelke		selu	Podding	Vijay
61	Babasaheb Hudage		Nilanga	Podding	Virat

62	Ravasaheb Patil	Osmanabad	Garsoli	Podding	Virat
63	Shakarrao Sabade		Renapur	podding	Vijay
64	Sadashiv futane		Beldar pimpri	Flowering	Vijay
65	Mamasaheb futane		Beldar pimpi	Flowering	Vijay
66	Vishal Bhore		Kalamb	Podding	Vijay
67	Sudhir Sagare		Dhoki	Podding	Vijay
68	Mahadeo Rote		Tuljapur	Podding	Vijay
69	Bhanudas Tate	Hingoli	Savad	Podding	Virat
70	Vinod Tate		Savad	Podding	Virat
71	Gajanan Tate		Savad	Podding	Local
72	Rohidas Mhase		Hingoli	Podding	Local
73	Ramesh Deokate		Hingoli	Podding	Vijay

3.2.2 Isolation, Pathogenicity, Reisolation, Symptomatology

3.2.2.1 Isolation

Chickpea plants, showing typical dry root rot symptoms were collected from Agricultural Research station Badnapur. The roots of infected plants were washed in tap water to remove soil adhered to roots. Root of the diseased plant was cut in small pieces before isolation to avoid contamination. The roots were cut into small bits (2.5mm) with sterilized blade. These bits were then disinfected with 0.1 per cent aqueous solution of mercuric chloride (HgCl_2) for two minutes and washed with three subsequent changes of sterilised water to remove traces of mercuric chloride. These bits were blot dried and four bits per petriplate were placed on the solidified potato dextrose agar (PDA) plates. These plates were then incubated at $28 \pm 2^\circ\text{C}$ temperature for seven days. The well isolated fungal growth, free from any contamination was transferred to agar slant by hyphal tip method. By frequent sub culturing, it was purified and agar slants showing pure fungal growth were maintained and used for further studies.

3.2.2.2. Pathogenicity

The full grown inoculum of each isolate of *R.bataticola* multiplied on sand maize medium was mixed @50 g/kg soil in each sterilized

earthen pot of 18 cm diameter. Three pots were used for each isolate as three replications. Control without any inoculums was kept for comparison. Each inoculated pot was moistened with water and covered with polythene sheet and incubated for 15 days. The surface sterilized (with 2.5 per cent NaOCl₂) seeds of chickpea variety JG- 62 were sown @10 seeds per pot. The pots were kept in glass house. Observations were recorded on root rot incidence.

$$\text{Per cent disease incidence (PDI)} = \frac{\text{Number of infected plants}}{\text{Number of plants inoculated}} \times 100$$

3.2.2.3. Reisolation

The reisolutions were made on PDA plates from roots of artificially diseased plants showing typical symptoms of dry root rot. The fungal growth obtained was transferred on potato dextrose agar slants for comparing with original culture.

3.2.2.4. Symptomatology

The symptoms of dry root rot were observed at flowering and pod filling stage till complete drying of plants in artificially inoculated soil and under field conditions.

3.2.3. Morphological studies

Morphological characters of the fungus were studied using fresh culture grown on PDA plates and preparing slides for these cultures for observations under microscope. Morphological character viz., size, shape, and septation were recorded from seven days old culture (30 days old for chlamydospores) grown on PDA at 27±1° C. The conidial spores' measurements were recorded with the help of Filar micrometer.

3.2.4. Identification of pathogen

Based on morphological characteristics, symptomatology and pathogenicity test, the test fungus was identified as *Rhizoctonia bataticola* (Taub) Butler.

3.3 Disease management

3.3.1 *In-vitro* studies

3.3.1.1 Evaluation of fungicides against *R. bataticola*

The effect of fungicides on *R. bataticola* (Taub) Butler was evaluated *in-vitro* by employing “Poisoned food techniques” (Nene and Thapliyal, 1993). The requisite amount of each fungicide based on active ingredient was added in an autoclaved and cooled (45°C) Potato dextrose agar medium to obtain desired concentrations. The fungicides used with their concentration were Carbendazim (0.2 per cent), Captan (0.3 per cent), Thiram (0.3 per cent), Mancozeb (0.3 per cent), Carbendazim + Thiram (2:1) (0.3 per cent), Carbendazim + mancozeb (1:2) (0.3 per cent), copper oxy chloride (0.3 per cent), Zineb. (0.3 per cent), Plain PDA without fungicide served as control. The PDA was then poured into sterile glass petriplates (90 mm diameter) and allowed to solidify. These plates were inoculated with a 5 mm mycelial disc of the test fungus and incubated for 7 days at 28 ± 2°C. All the treatments were replicated thrice. Observations on radial mycelial growth in treatment plates were recorded after 7-8 days. Per cent inhibition of the test fungus over control was calculated by applying the formula given below (Vincent, 1927).

$$\text{Per cent inhibition (I)} = \frac{C - T}{C} \times 100$$

Where,

C = Growth (mm) of test fungus in control plate.

T = Growth (mm) of test fungus in treatment plate.

3.3.1.2 Evaluation of botanicals (Plant extract)

The effect of phytoextracts of various plant species were tested in vitro by poisoned food technique to know their inhibitory effect on growth of *R. bataticola*. Healthy fresh plant parts i.e. leaves, bulbs or rhizomes were taken, washed thoroughly with fresh water and finally rinsed with sterilized distilled water. Fifty gram of plant parts were cut into small pieces and minced with the help of grinder by adding 50 ml sterilized distilled water. The phyto extract were filtered through double layered muslin cloth in 150 ml conical flasks and plugged with non-absorbent cotton. These filtered extract were autoclaved at 1.2 kg /cm² pressure for 20 minutes. Autoclaved extract were added into PDA. Plain PDA without plant extract served as control. The PDA was then poured into sterile glass petriplates (90 mm diameter) and allowed to solidify. The petriplates were inoculated aseptically after solidification by placing 5 mm mycelial disc at the centre, cut aseptically with cork borer from 10 days old pure culture of *R. bataticola*. Three replication of each treatment were maintained. The plates without phytoextract served as control. The petriplate were incubated at 27 ± 2° C temperature till the complete coverage in controlled plate. The percent growth inhibition (PGI) of the pathogen was worked out by using formula given by Vincent (1947)

$$\text{Per cent inhibition (I)} = \frac{C - T}{C} \times 100$$

Where,

C = Growth (mm) of test fungus in control plate.

T = Growth (mm) of test fungus in treatment plate

3.3.1.3 Evaluation of Biocontrol agents against *R. bataticola*

Eight biocontrol agent viz., *T. viride*, *T. harzianum*, *T. hamatum*, *T. longibrachiatum*, *T. koningii*, *Aspergillus niger*, *Aspergillus flavus*, *Pseudomonas flurescens* were assessed against *R. bataticola* by dual culture method (Upadhyay and Rai, 1987). Mycelial disc of 5 mm diameter were cut from the margin of 7 days old culture of both test pathogen and placed opposite to each other on PDA in petriplate. The inoculums discs were placed 30 mm away from each other. The petriplate inoculated with disc of *Rhizoctonia* alone served as control. The inoculated plates were incubated at $27 \pm 2^{\circ}\text{C}$ in BOD incubator for 7 days. The percent growth inhibition of inserting colonies was calculated as per formula given below.

$$\text{Per cent inhibition (I)} = \frac{A_1 - A_2}{A_1} \times 100$$

Where,

A_1 = Area covered by pathogen in control plate.

A_2 = Area covered by pathogen in dual culture plate

3.3.2 Pot culture experiment

3.3.2.1 Evaluation of fungicide as seed dresser against *R. bataticola*

Efficacy of all eight fungicides was tested in pot culture under glass house conditions. Chickpea seeds of cultivar JG-62 were treated with fungicide at 1 g/kg or 2 g/kg or 3 g/kg seed. The treated and untreated seeds (control) were sown in artificially inoculated soil at 10 seeds per pot. Each pot (18cm diameter) was surface sterilized in 0.1 per cent formalin and filled with potting mixture and mixed with inoculums @ 20 g/kg soil. The pots were moistened, covered with polythene sheet and incubated for 15 days. Each treatment was replicated three times. Percent disease incidence (PDI) was

recorded at 30, 45,60,75,90 days after sowing. The percent disease incidence over control was calculated.

$$(PDI) = \frac{\text{DI in untreated seeds} - \text{DI in treated seeds}}{\text{DI in untreated seeds}} \times 100$$

3.3.2.2 Evaluation of Bio agents as seed dresser against *R. bataticola*

Efficacy of all eight bioagents was tested in pot culture under glass house condition .Chickpea seeds of cultivar JG -62 were treated with bioagent 1.5g/kg seed. The treated and untreated seed (control) were sown in artificially inoculated soil at 10 seed per pot. Each pot was surface sterilized in 0.1 per cent formalin and filled with potting mixture with inoculums @20g/kg soil. The pot were moistened covered with polythene sheet and incubate for 15 days is treatment was replicated three times. Percent disease incidence was recorded at 30, 45,60,75,90 days after sowing .The percent disease incidence over control was calculated.

$$(PDI) = \frac{\text{DI in untreated seeds} - \text{DI in treated seeds}}{\text{DI in untreated seeds}} \times 100$$

3.3.2.3 Evaluation of Botanicals (plant extract) as seed dresser against *R. bataticola*

Efficacy of all eight botanicals (plant extract) was tested in pot culture under glass house condition. Chickpea seed of cultivar JG-62 were treated with plant extract. Leaves, bulb, rhizomes cleaned by tap water. Plant extract was prepared by crushing leaves or bulb or rhizomes (1:1(w/v)). Extract filtered through muslin cloth and soaked seeds in plant extract for 2 hour. After this, seed sown in pots containing sick soil. Percent disease incidence was recorded at 30,45,60,75 and 90 DAS. The percent disease incidence over control was calculated as:

J-7 666

$$(PDI) = \frac{\text{DI in untreated seeds} - \text{DI in treated seeds}}{\text{DI in untreated seeds}} \times 100$$

3.4. Screening technique

Two test entries alternated with susceptible check JG-62 were planted in two replications. The screening was done in sick plot. The observation viz., total number of plants, total number of infected plants and the average of dry root rot in each entry was calculated. The observations on dry root rot were recorded in percentage of disease incidence.

Based on the disease incidence, test lines are categorized for their reaction to Dry root rot as per rating scale given by Nene (1981).

Disease rating scale:

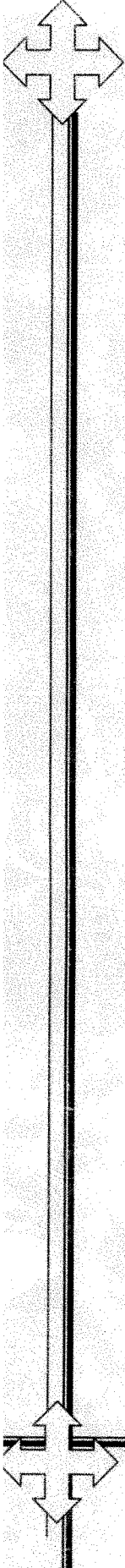
Table: Disease reaction category based on disease incidence

Disease incidence (%)	Disease reaction
1.0-10.0 %	Resistant
10.1-20.0%	Moderately resistant
20.1-30.0%	Moderately susceptible
30.1-50.0%	Susceptible
Above 50 %	Highly susceptible

3.5. Statistical analysis

The experimental data was statistically analyzed at computer centre, College of Agriculture, Badnapur. Observation of per cent disease incidence and intensity were converted in value of arcsine transformation for statistical analysis.





RESULTS AND DISCUSSION

CHAPTER-IV

RESULTS AND DISSCUTION

Present studies on the dry root rot of chickpea caused by *Rhizoctonia bataticola* were undertaken during *Rabi* 2014 with aspects viz., survey, isolation, pathogenicity test, *in vitro* and pot culture evaluation of fungicides, botanicals and bioagents, screening of various germplasm. The results obtained on all these aspects are presented in the following paragraphs.

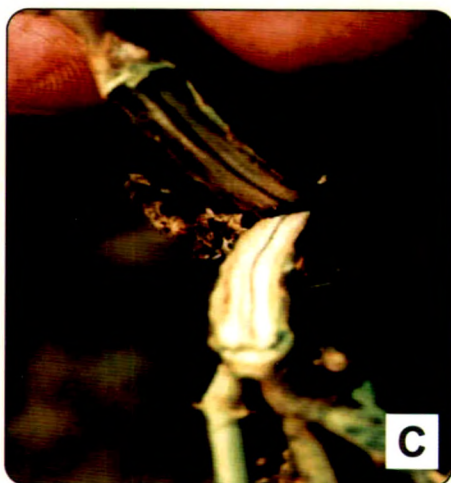
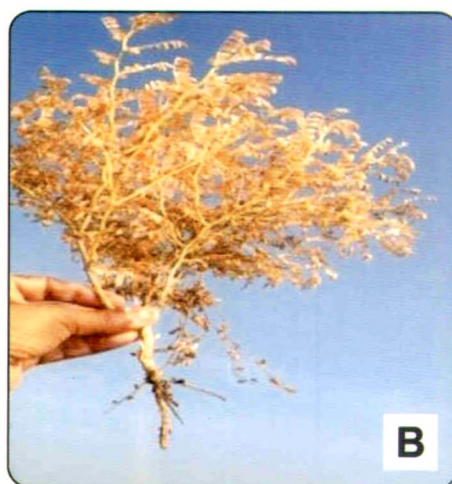
4.1 Survey of Dry root rot in Marathawada region

During *Rabi*, 2014-15 a roving survey was conducted during the months of December (2014) to February (2015) in Marathawada region by selecting Chickpea fields adjacent to road, approximately at every 15-20 km distance.

4.1.1 Dry root rot incidence in Marathawada region

Results (Table 1) indicated that the dry root rot average incidence in the Marathawada region (8 districts) during *Rabi*, 2014 was ranged from 10.12 to 23.14 per cent. However, the highest average incidence was recorded at Latur district (23.14 per cent) followed by Jalna (22.2 per cent) and moderate disease incidence was recorded in Parbhani, Beed, Nanded districts. Whereas lowest dry root rot incidence was recorded in Aurangabad district (10.12 per cent). Mean of average percentage of Dry root rot disease incidence of Marathawada region was 15.37 per cent.

The perusal of the result indicate that the average dry root rot incidence in Latur was found to be maximum (23.14 per cent). This may be due to prolong dry period and low rainfall. Prominently, crop at pod filling



**A, B, C - Chickpea plant showing typical symptoms of Dry root rot
D,E - Dry root rot symptoms 45 DAS and 65 DAS**

Plate 1. Typical symptoms of Dry root rot of Chickpea

stage caught in moisture stress. These observations are in conformity with the following earlier workers.

The survey was carried out for recording the incidence of dry root rot of chickpea. It was observed the yield of chickpea seed was adversely affected culminating into a total loss. Dry root rot of chickpea crop induced by *R.bataticola* was the most serious disease prevalent all over the country (Saksena *et al.*, 1979).

Prajapati *et al.* (2004), Conducted survey on incidence of Chickpea dry root rot in Uttar Pradesh. Survey was conducted at 20 locations at different agro-climatic regions of Uttar Pradesh. The disease was wide spread in occurrence in UP with 19.0 to 42.0 per cent incidence.

Kohire *et al.* (2006), Carried out a survey in 2004-2006. They reported that wilt, dry root rot and stunt were associated with infected chickpea in Marathawada region of Maharashtra. Disease incidence varied from 1.6 to 16.5 per cent. Maximum average incidence was observed in Parbhani (16.5 per cent).

The dry root rot disease was prevalent in all pea growing areas of state during 2006-2009. The incidence of wilt and root rot was 20 per cent in temperate wet region, 15per cent in sub temperate region , 12 per cent in temperate dry region and 11per cent in subtropical region. (Rana *et al.* (2009).

Table- 1. Survey of dry root rot incidence in Marathawada region during Rabi-2014-15

Sr. No	District	No of location Traversed	Location	Av % of dry root rot disease incidence in each Districts
1	Aurangabad	8	Shivoor, Devgoan, Kolhi, Ganori, Pathri, Kannad, Gangapur, Panwadod	10.12
2	Parbhani	4	Rawallgoan, Purna, Jintur, Parbhani,	17.6
3	Latur	7	Nilanga, Arvi, Murud, Garsoli, Renapur	23.14
4	Beed	7	Pimpri ,Ghumari, Ashti, Neknur, Kalasambar, Kaij, Ambajogai	15.20
5	Jalna	8	Hanamantkheda, Deulgaon, Khasgaon, Gevrai bazaar, kandari, Nikalaj, Rajur, Badnapur	22.2
6	Hingoli	2	Savad, Hingoli	12.4
7	Osmanabad	4	Beldar Pimpri, Kalamb, Dhoki, Tuljapur	11.16
8	Nanded	3	Pala, Ganjgaon, Sagaroli	14.23
Average Dry root rot percentage in Marathawada region				15.75

Av % of dry root rot disease incidence in each Districts
(2014-15)

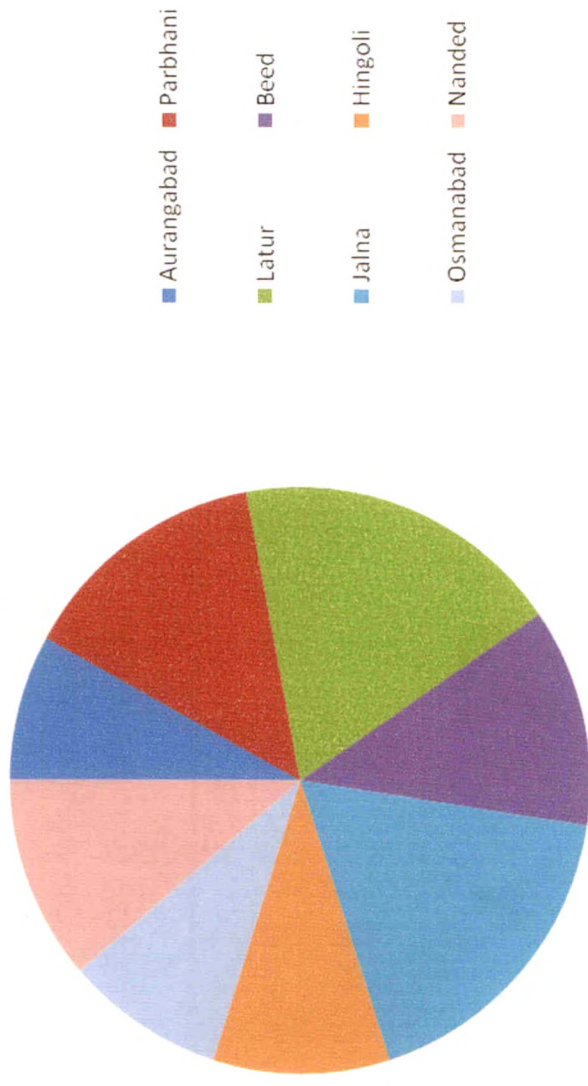


Fig 1. Average dry root rot incidence in each district of Marathawada region (2014-15)

4.2 Isolation

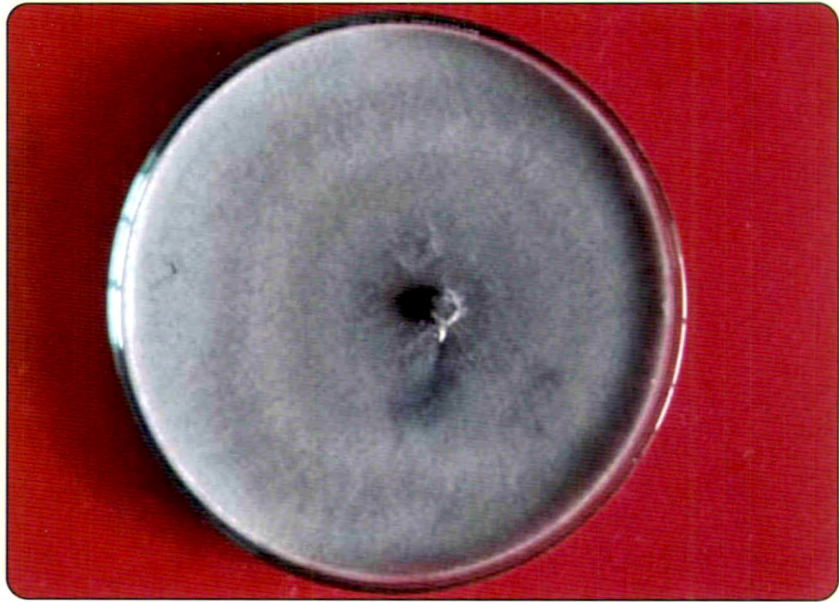
The test pathogen (*Rhizoctonia bataticola*) was isolated successfully on the basal culture medium potato dextrose agar, from the roots showing typical symptoms of dry root rot by standard tissue isolation method (PLATE –II). The pathogen produced dark brown to black colored colonies to septate mycelial growth on Potato dextrose agar (PLATE -II). Pure culture of *Rhizoctonia bataticola* was obtained by sub culturing on PDA plates and maintained on PDA slant for further studies.

Earlier isolation of *R. bataticola* from infected chickpea plant was carried out by Mitra (1931), Sharma and Khare (1969), Bhatti *et al.* (1987), Beniwal *et al.* (1992), Demirci *et al.* (1998), Rangaswamy and Mahadevan, 1999, Sharma *et al.* (2004).

4.3 Pathogenicity test:

The pathogenicity test was carried out by soil inoculation method. Typical dry root rot symptoms showing drying of plant started to develop 30 – 40 days after inoculation, indicating thereby that isolate was pathogenic to chickpea cultivar JG-62. In the advanced stage of disease, the plant started drying from the base to top and at flowering stage. It had suddenly and completely dried. The affected plants could be easily pulled out as there was necrosis of roots. The bark near ground level got shredded. The blackish sclerotial bodies were observed on the exposed root, particularly at collar region. Symptoms developed were found to be similar to that of naturally infected chickpea plants (Plate-III).

The pathogenicity of the isolates tested by soil inoculation method indicates that the variety JG-62 caused 100 per cent disease incidence than



Pure culture of *R. bataticola* (Mycelial stage)



Healthy Plant

Diseased Plant

Plate 2. Pure culture of *R. bataticola* (Mycelial stage) and pathogenicity

other varieties. This may be due to the difference in ability of isolate to colonize the epidermal and cortical cells of host and production of peptic enzyme involved in cell degradation under different environmental conditions. Earlier, Mitra (1931) proved pathogenicity of *R. bataticola* isolated from chickpea on potato. Bhatti *et al.* (1987), Beniwal *et al.* (1992) and Katariya *et al.* (2007) proved pathogenicity of *Rhizoctonia bataticola* on chickpea.

4.4. Reisolation

The fungus was reisolated on PDA from artificially inoculated plants showing typical dry root rot symptoms and the fungus growth was similar to original culture.

4.5. *In vitro* evaluation of fungicides against *R. bataticola*

Observations in respect of effect of different concentrations of fungicides on inhibition of mycelial growth of pathogen on PDA are depicted in table 2 which revealed that the growth of *R. bataticola* differed significantly due to fungicides at two concentrations. The maximum mycelia growth inhibition was recorded in Carbendazim + Mancozeb (100 per cent) which was followed by Thiram + Carbendazim (99.22 per cent). Next to these treatment Carbendazim recorded 97.74 per cent inhibition which is followed by captan (90.97 per cent inhibition) and Mancozeb (90.55 per cent inhibition), Thiram was recorded 78.21 per cent mycelial inhibition which is followed by Copper oxychloride (40.60 per cent inhibition) and minimum percent inhibition recorded by Zineb (32.42 per cent).

Interaction effect of fungicides and their concentration significantly affected the mycelial growth *in vitro*. The interaction of fungicide Carbendazim + Mancozeb @ 1000 ppm concentration inhibit the mycelial

growth 100% more significantly over other interaction over other interaction which is followed by Thiram + Carbendazim (98.44 per cent). Same pattern of interaction was observed at 2000 ppm concentration.

The efficacy of Carbendazim against *R.bataticola* in present studies is in confirmation with the finding of earlier workers such as Rathore (2001) reported that seed treatment with Bavistin 50 WP recorded minimum percent disease intensity and maximum seed yield followed Captan 75 SD against dry root rot of cluster bean.

Prajapati *et al.* (2002) reported that Carbendazim 25 DS, Carbendazim 50WP+thiram 80 WP were completely inhibited the growth of *R.bataticola* and were found to be most effective.

Sharma and Gupta (2004) studied efficacy of the fungicide in the control of Chickpea dry root rot caused by *Rhizoctonia bataticola* and reported that seed treatment with Carbendazim (1 gm /kg seed) showed the minimum incidence of disease (17-21per cent) and increased grain.

Vijay –Mohan *et al.*(2006) studied the fungicidal management of dry root rot disease and yield of chickpea and reported that Carbendazim (0.2per cent) and Etaconazole (0.1per cent) used as seed treatment ,soil drenching and combined effect recorded lowest disease incidence of 15.6 and 18.2 per cent and highest grain yield of 192 and 18.9 q/ha respectively.

Kaushal and Sood (2008) studied management of root rot in chickpea and reported that seed treatment with Thirum and Bavistin @ 2.0gm/kg seed gave good control of root rot of chickpea under field condition.

Also the efficacy of Carbendazim against *R.bataticola* in present studies is in confirmation with the finding of Goyal and Mehrotra (1981) and Taneja and Grover (1982), Sarwar and Gopal Raju (1985), Ramadoss and Shivprakasan (1987), Chauhan *et al.*(1988), Singh and Sindan (1988), Peshney *et al.*(1992), Rathore (2001), Prajapati *et al.*(2004), Sandipan *et al.*(2007).

Table 2. *In Vitro* evaluation of fungicides against *R.bataticola*

Sr. No.	Treatments	Growth of fungus @ 1000 ppm fungicide (mm) *	% inhibition over control	Growth of fungus @ 2000 ppm fungicide (mm) *	% inhibition over control	Mean % inhibition over control
1	Thiram	20.50	75.78	16.40	80.64	78.21
2	Carbendazim	2.19	97.41	1.63	98.07	97.74
3	Captan	9.13	89.21	6.15	92.74	90.98
4	Thiram +Carbendazim	1.32	98.44	0.00	100	99.22
5	Mancozeb	9.35	88.95	6.65	92.15	90.55
6	Carbendazim +Mancozeb	0.00	100	0.00	100	100
7	Copper oxychloride	52.54	37.71	48.73	42.50	40.60
8	Zineb	62.18	26.56	52.30	38.58	32.42
9	Control	84.67		84.75		
	SE. ±	0.86		0.91		
	CD @ 1%	2.30		2.94		

* Mean of three replications

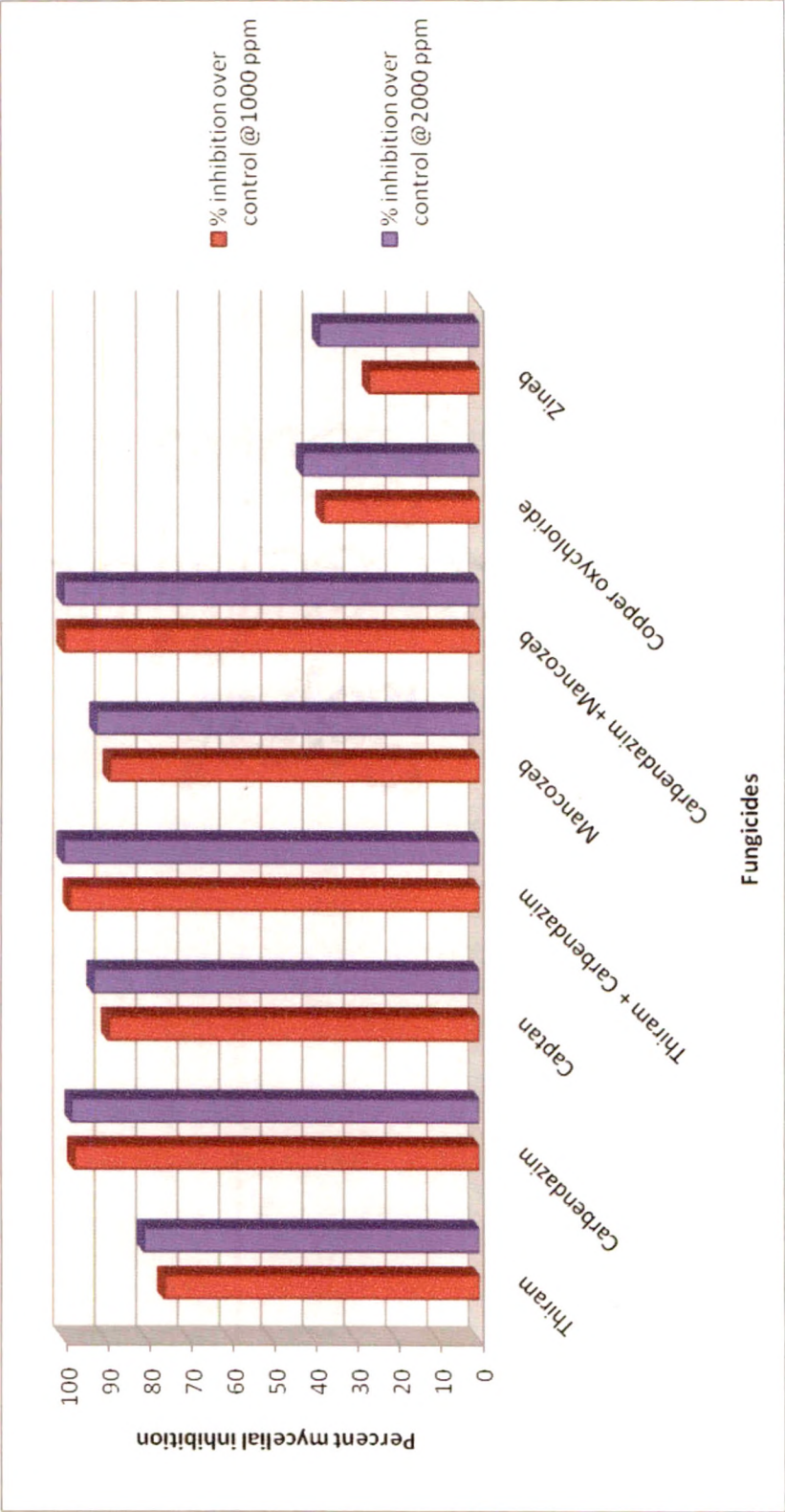
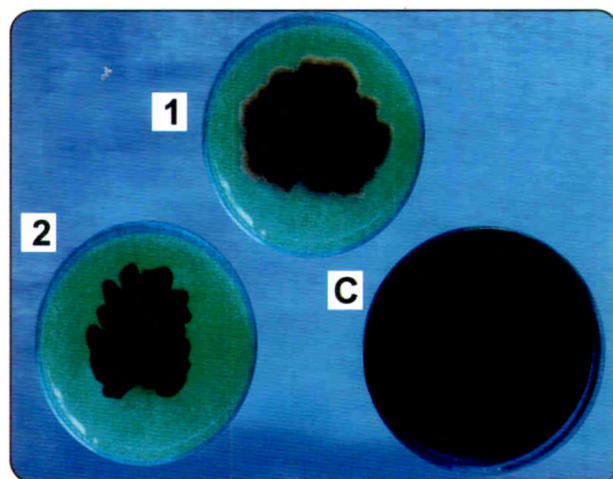
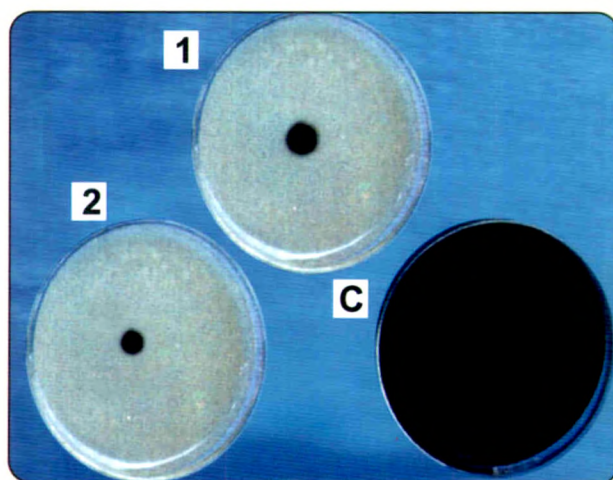


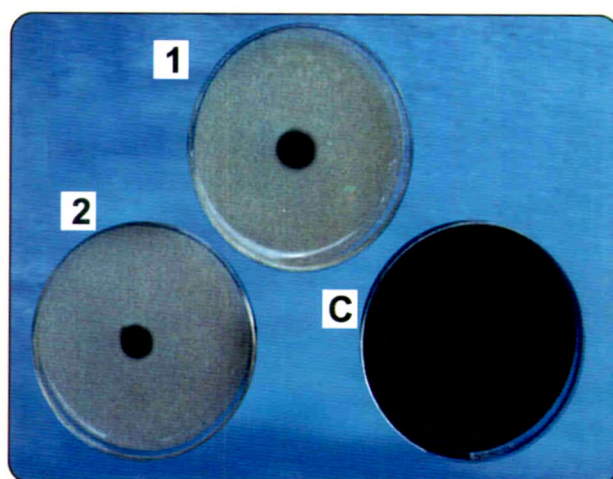
Fig 2 : Effect of fungicides on mycelial inhibition of *R. bataticola* at different concentration



Thiram

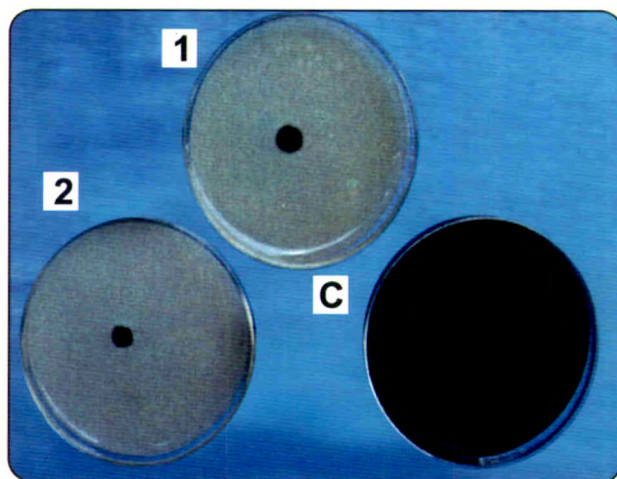


Carbendazim

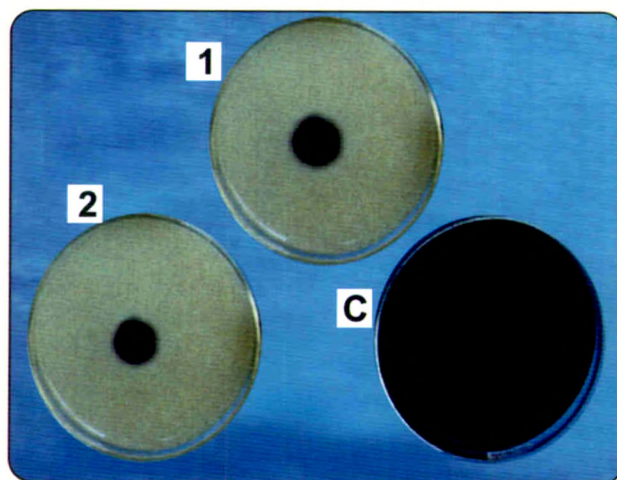


Captan

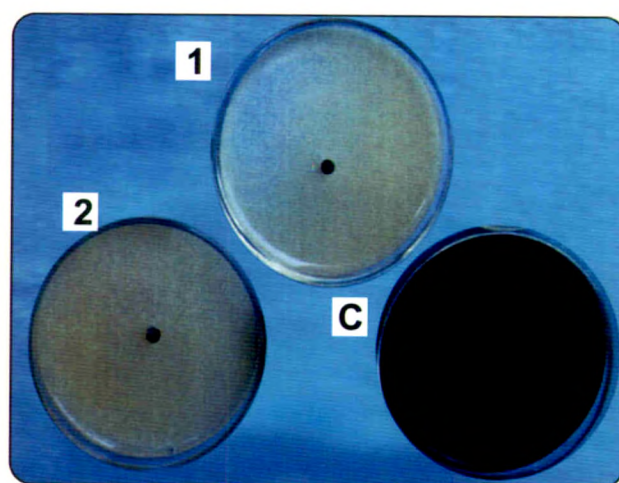
Plate 3. Effect of fungicide on mycelial growth of *R. bataticola*



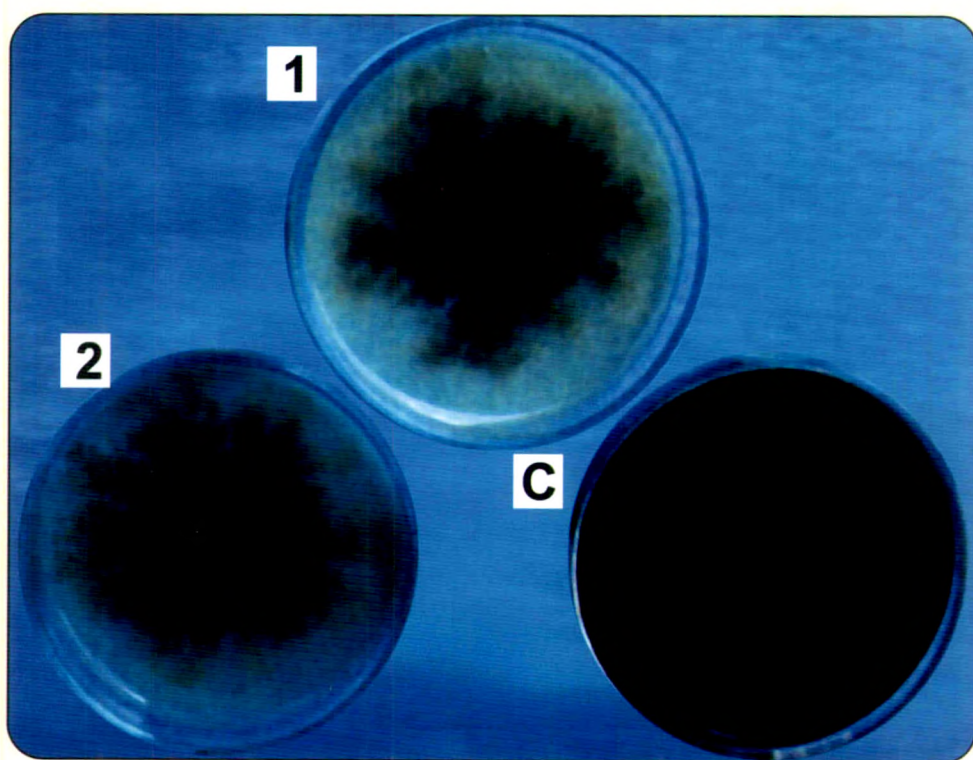
Thiram + Carbendazim



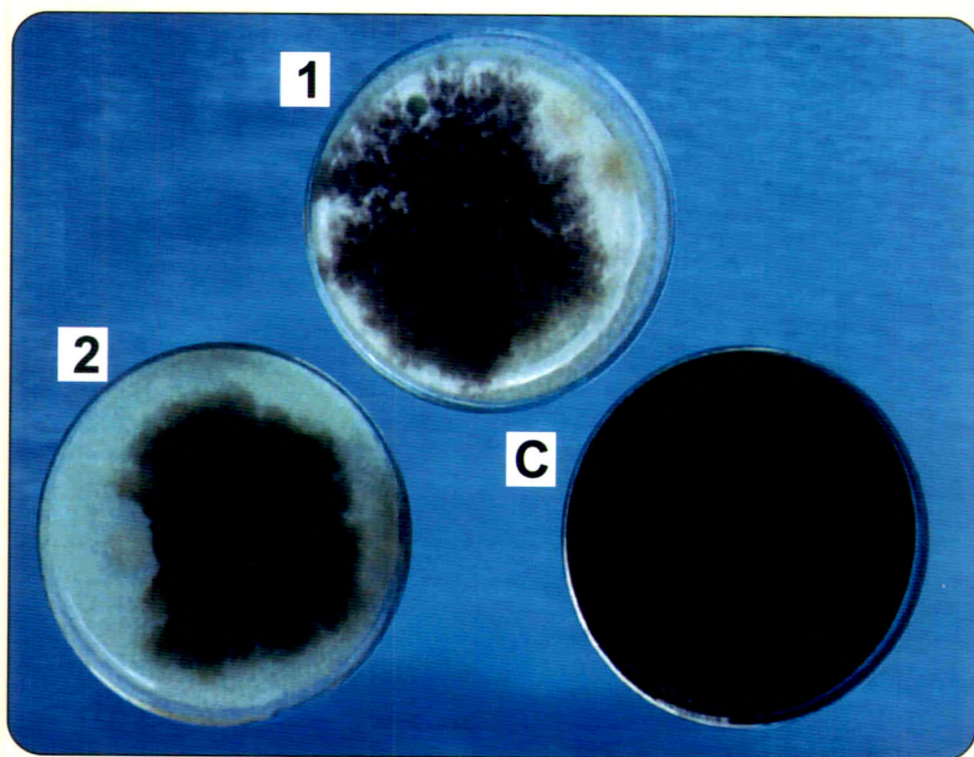
Mancozeb



Carbendazim + Mancozeb



Copper oxychloride



Zineb

4.6. *In vitro* evaluation of plant extracts

Those plant extract found most effective against various soil born diseases which were selected and evaluated against dry root rot of chickpea. Observation in respect of various botanicals on inhibition of mycelial growth of pathogen on PDA are depicted in Table 3. The result presented in table revealed that all the plant extracts inhibited the growth of fungus as compared to control. The clove extract of Garlic (*Allium sativum*) allowed 22.03 mm growth of pathogen which is followed by finger extract of Turmeric (*Curcuma longa* L.) (32.05 mm), rhizome extract of Ginger (*Zingiber officinale* L.) (34.93 mm). The next in order of merit were leaves extract of Mehandi (39.67mm), leaves extract of Bougainvillia (45.41 mm). Leaves extract of Parthenium (58.34mm). Whereas leaves extract of Lantana (*Lantana camera*) (83.53 mm) is least effective.

The clove extract of Garlic (*Allium sativum*) (75.55 per cent) produced maximum L.) (63.98per cent) inhibition to growth of pathogen. The next in order of merit were leaves extract of Mehandi (55.42 per cent) leaves extract of Bougainvillia (48.97per cent), leaves extract of Parthenium (34.44 per cent). Whereas leaves extract of Lantana (*Lantana camera*) (6.14 per cent) is least effective in inhibiting the growth of pathogen.

The present studies are in agreement with those described earlier by other workers viz., Datar (1999) studied the effect of botanicals on *R.bataticola*. Dubey and Dwivedi (1991) found fungi toxic properties of *A.sativum* against sclerotial viability of *R.bataticola*. Kane *et al.* (2002) reported that crude extract of *Allium sativum* and *Zingiber officinale* were effective in inhibiting the mycelial growth of *R.bataticola* to the extent at 100 percent. Tandel *et al.* (2010) tried phyto-extract of eleven plant species against *M.phaseolina* of Green gram and revealed that the onion bulb extract produced

Table 3. *In vitro* evaluation of Botanicals against *Rhizoctonia bataticola*

Sr. No.	Name of plant	Botanical name	Average colony (mm) *	% growth inhibition over control
1	Garlic	<i>Allium sativum</i>	22.03	75.25
2	Turmeric	<i>Curcuma longa</i>	32.05	63.98
3	Ginger	<i>Zingiber officinale</i>	34.93	60.75
4	Neem	<i>Azadiracta indica</i>	82.71	7.07
5	Lantana	<i>Lantana camera</i>	83.53	6.14
6	Mehandi	<i>Lawsonia inermis</i>	39.67	55.42
7	Parthenium	<i>Patrhenium hysterochorus</i>	58.34	34.44
8	Bougainvillea	<i>Bougainvillea spp.</i>	45.41	48.87
9	Control		89.00	
S.E. ±			1.41	
CD at 1%			4.16	

* Mean of three replications

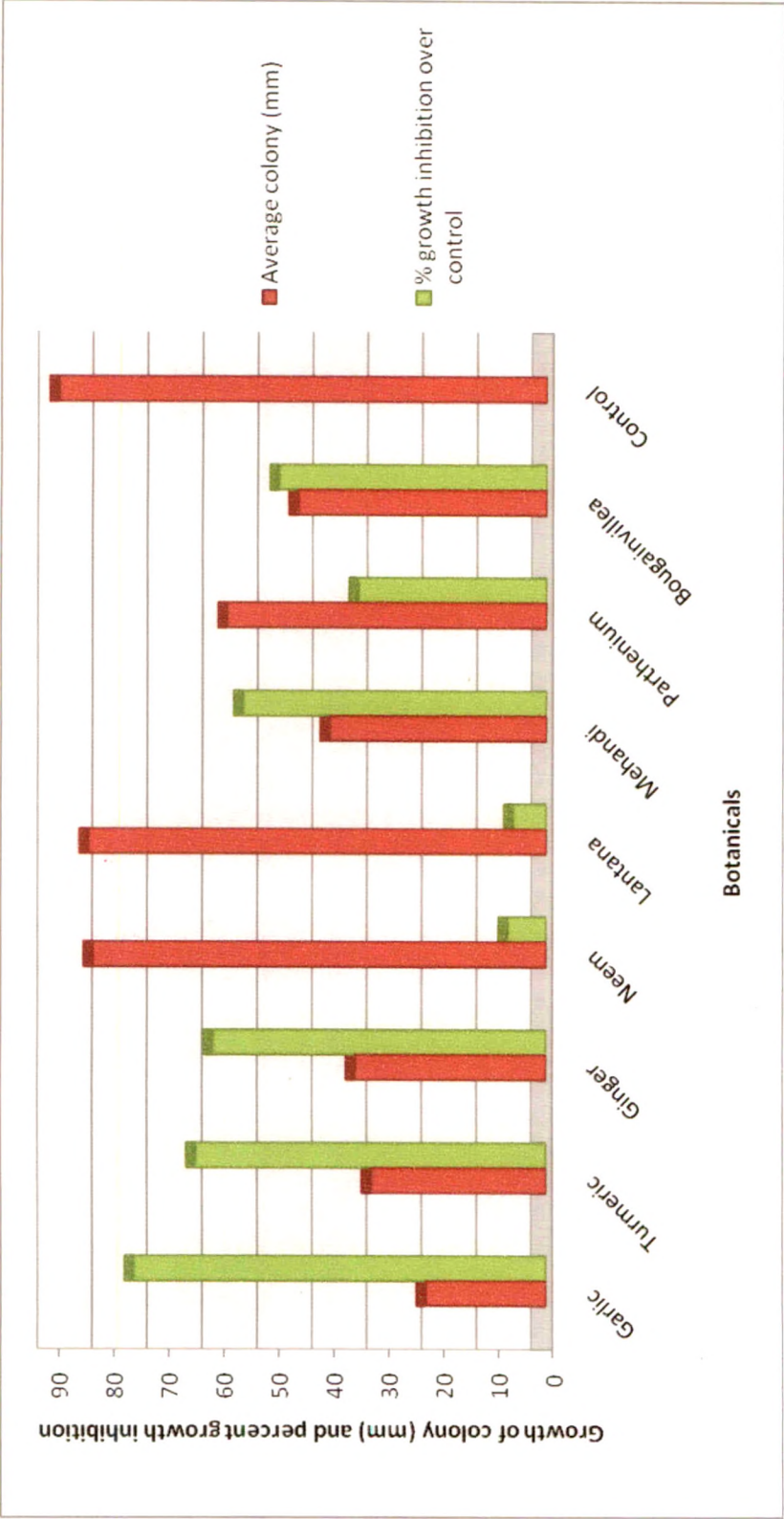
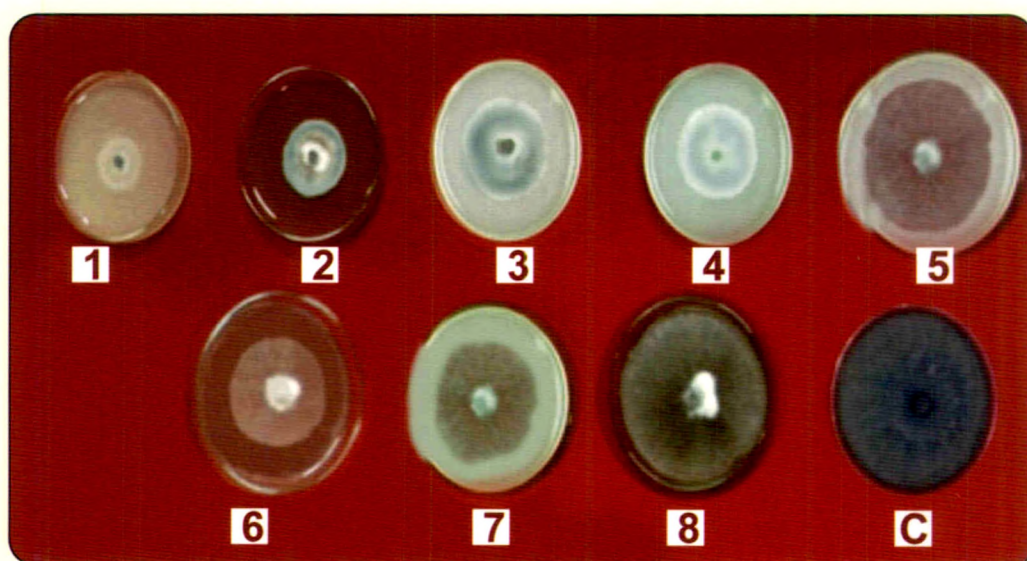


Fig 3: Effect of botanicals on mycelial growth and percent inhibition of *R. bataticola*



1. Garlic
2. Turmeric
3. Ginger
4. Neem

5. Lantana
6. Mehendi
7. Parthenium
8. Bougainvillea
C. Control

Plate 4. Effect of botanicals on mycelial growth of *R. bataticola*

maximum inhibition (98.14 per cent) which is followed by extract of Ginger, Neem, Garlic.

From this study it is clear that Garlic and Turmeric were found high effective in reducing the growth of *R.bataticola* causing dry root rot of chickpea. This may be due to presence of curcumin in Turmeric and allicin in Garlic.

4.7 *In vitro* evaluation of bioagents against *Rhizoctonia bataticola*

All the antagonists showed significant reduction in mycelial growth of *Rhizoctonia bataticola* when compared to control. The result presented in Table 4 reveal that all the bioagents suppress the colony growth of *R.bataticola* . An average growth of *Rhizoctonia bataticola* colony with *T. viride* is 10.41mm and an average growth of *R.bataticola* colony with *T. harzanium* is 30.61 mm which is followed by *T.hamatum* (34.60 mm), *T.longibrachiatum* (38.12mm), *T.koningii* (74.63 mm), *A.niger* (40.81 mm), *A.flavus* (41.58 mm), *P. fleurescens* (50.19 mm).

The data pertaining to percent inhibition of mycelial growth of *R.bataticola* due to antagonists are presented in Table-4. The suppression growth of pathogen was maximum with *T. viride* (80.05 per cent) which is followed by *T. harzanium* (58.98 per cent) and the least effective bioagents was *P. fleurescens* (32.75 per cent).

T. viride showed best performance against the pathogen followed by *T. harzanium*, *T.hamatum* , *T.longibrachiatum*, *A.niger*, *A.flavus*, *P. fleurescens* which also checked the fungus growth to some extent similar finding were also reported by Singh *et al.* (2006). Kaushal (2008) reported that the *T. harzanium* was effective in inhibiting the mycelial growth of *R.bataticola* the causal organism of chickpea dry root rot. Pan (2009) reported

Table 4 *In vitro* evaluation of bio agents against *R.bataicola*

Sr. No.	Treatment	Av. Growth of <i>R.bataicola</i> colony(mm) *	% Growth inhibition over control
1	<i>T. viride</i>	10.41	86.05
2	<i>T. harzanium</i>	30.61	58.98
3	<i>T.hamatum</i>	34.60	53.63
4	<i>T.longibrachiatum</i>	38.12	48.92
5	<i>T. koningii</i>	39.20	47.47
6	<i>A.niger</i>	40.81	45.32
7	<i>A.flavus</i>	41.58	44.28
8	<i>P. fleurescens</i>	50.16	32.75
9	Control	74.63	
S.E. ±		0.86	
CD at 1%		3.82	

* Mean of three replications

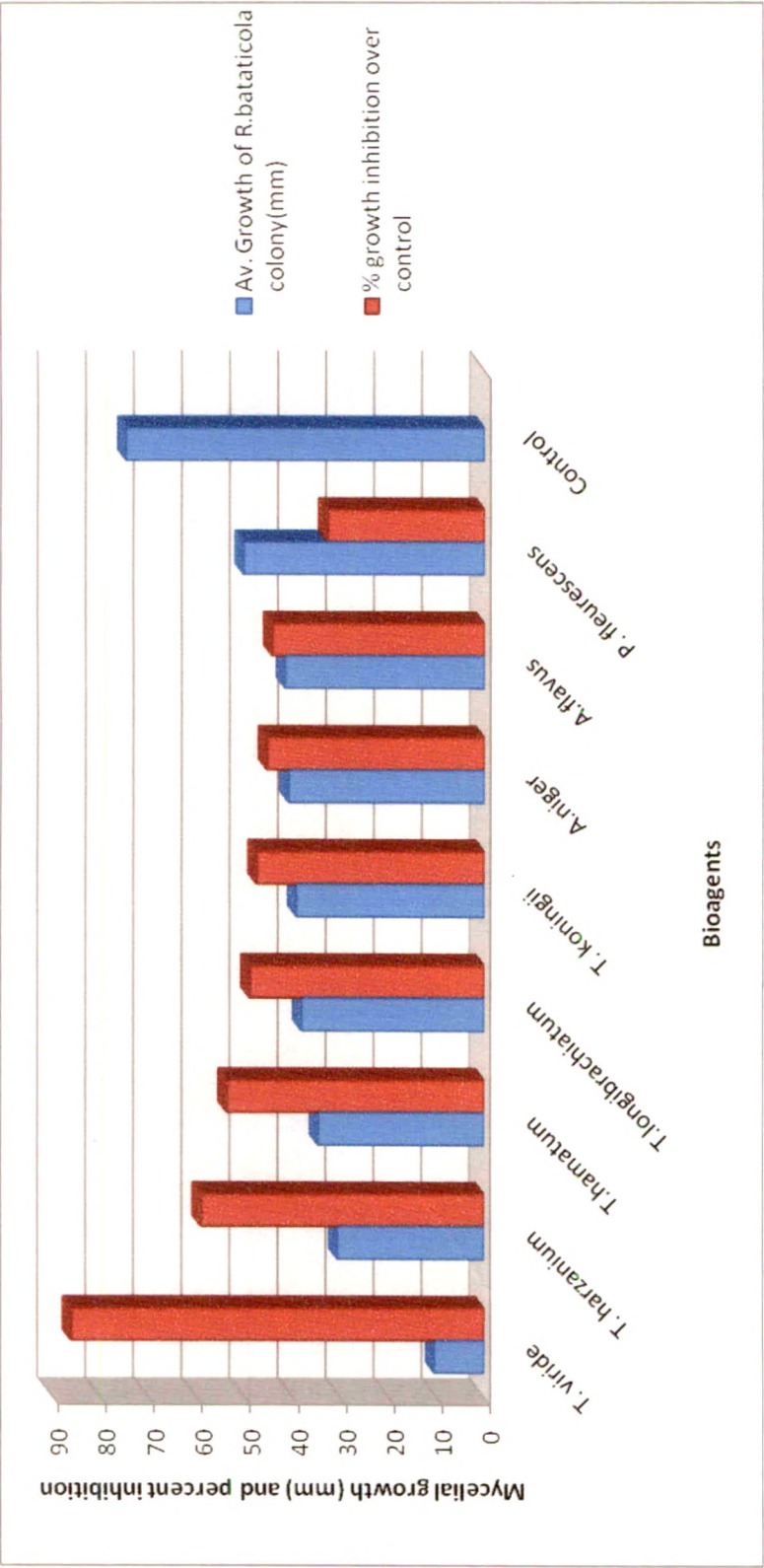
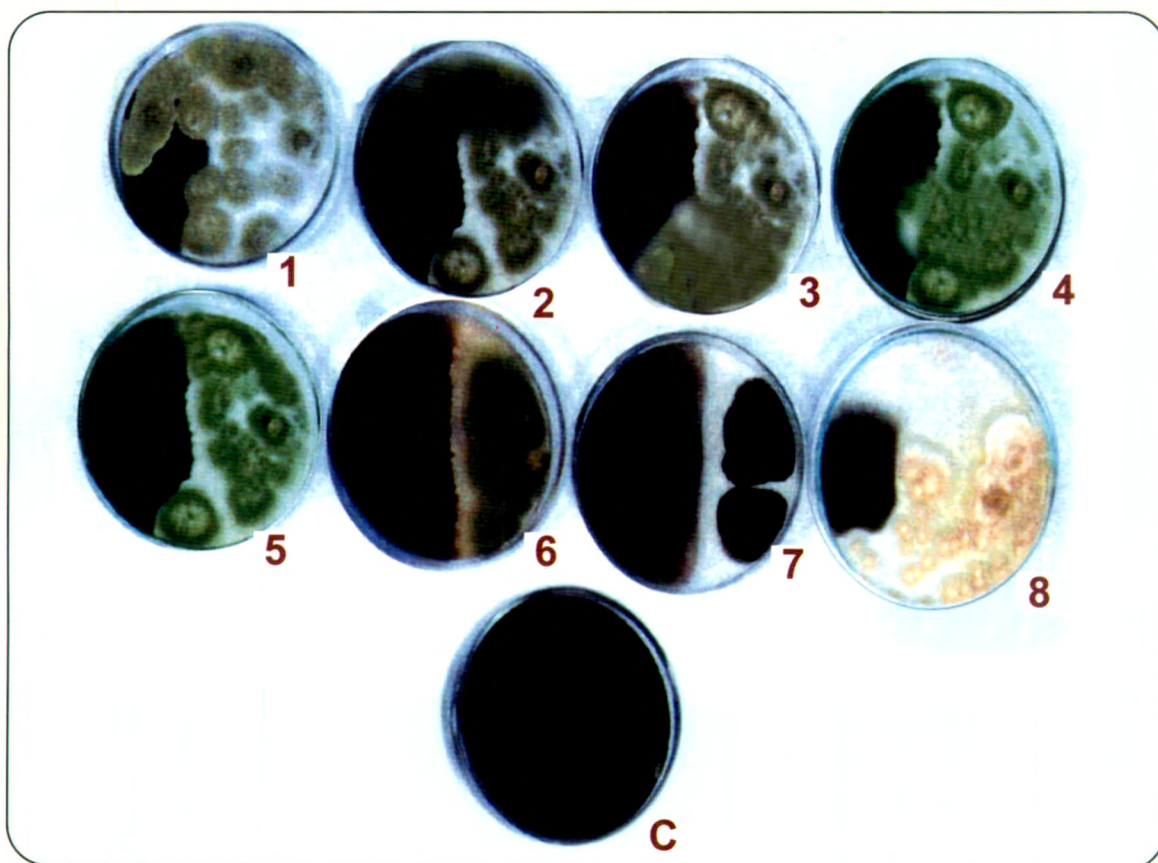


Fig 4: Effect of bioagents on mycelial growth and percent inhibition of *R. bataticola*



1. *T. viridi*
2. *T. harzianum*
3. *T. hamatum*
4. *T. longibrachiatum*

5. *T. Koningii*
6. *Aspergillus niger*
7. *Aspergillus flavus*
8. *Pseudomonas fluorescens*

Plate 5. Effect of fungicidal, botanical and bioagents seed treatment on incidence of Chickpea root rot in Pot culture

the antagonistic potential of *Trichoderma* isolates through production of volatile and non volatile substance against *M.phaseolina*. Choudhary (2010) tested bioagents viz., *T. viride* , *T. harzanium*, *A.vesicolor* for the control of dry root rot in Mungbean. In dual culture, *T. viride* , *T. harzanium*, *A.vesicolor* were effective in inhibiting the growth of *M.phaseolina* to an extent of 61-65 per cent.

4.7 Evaluation of fungicides, Bioagents and botanicals against *Rhizoctonia bataticola* with Chickpea cultivar JG-62 (Pot culture)

The three fungicides, three bio agents and two botanicals were evaluated by seed treatment method in Pot culture experiment with Chickpea cultivar JG-62 against *R. bataticola*. Observation on seed germination, percent disease incidence and percent inhibition over control were recorded.

Result (Table-5) revealed that fungicides, bio agents and botanicals were found effective against the test pathogen and significantly enhanced the seed germination and reduced the percent disease incidence in Chickpea over untreated control (Plate-V). In all treatment, maximum per cent germination was recorded in *T. viride* (93.33 per cent). This was followed by Carbendazim (90.00 per cent), Garlic (86.67 per cent), Thiram (83.33 per cent), *T.harzianum* (83.33 per cent), Captan (80.00 per cent), *T.hamatum* (80.00 per cent), Neem (70.00 per cent) as compared to untreated control (63.33 per cent).

The least disease incidence was recorded in treatment Carbendazim (11.56 per cent) and this was followed by Garlic (20.77 per cent), Thiram (26.42 per cent), Captan (28.29 per cent), *T. viride* (31.15 per cent), *T. harzianum* (34.81 per cent), *T .hamatum* (45.89 per cent), Neem (50.34 per cent) over control which showed maximum disease incidence (91.29 per cent).

Table-5 Evaluation of fungicides, Bioagents and botanicals against *Rhizoctonia bataticola* with Chickpea cultivar JG-62 (Pot culture)

Sr.No.	Treatments	Germination %	Mean disease incidence*	% inhibition over control
1	Thiram	83.33 (66.14)	26.42 (30.82)	71.09
2	Carbendazim	90.00 (73.65)	11.56 (19.85)	87.35
3	Captan	80.00 (63.43)	28.29 (32.06)	69.04
4	<i>T.viride</i>	93.33 (76.33)	31.15 (33.92)	65.91
5	<i>T.harzianum</i>	83.33 (66.14)	34.81 (36.13)	61.91
6	<i>T.hamatum</i>	80.00 (63.43)	45.89 (42.64)	49.75
7	Garlic	86.67 (68.86)	20.77 (26.66)	77.27
8	Neem	70.00 (56.79)	50.34 (45.19)	44.91
9	Control	63.33 (52.78)	91.39 (75.10)	00.00
	SE±	3.02	2.7	
	CD at 5%	9.5	8.7	

* Figures in parenthesis indicate arcsine transformed values * Mean of three replications

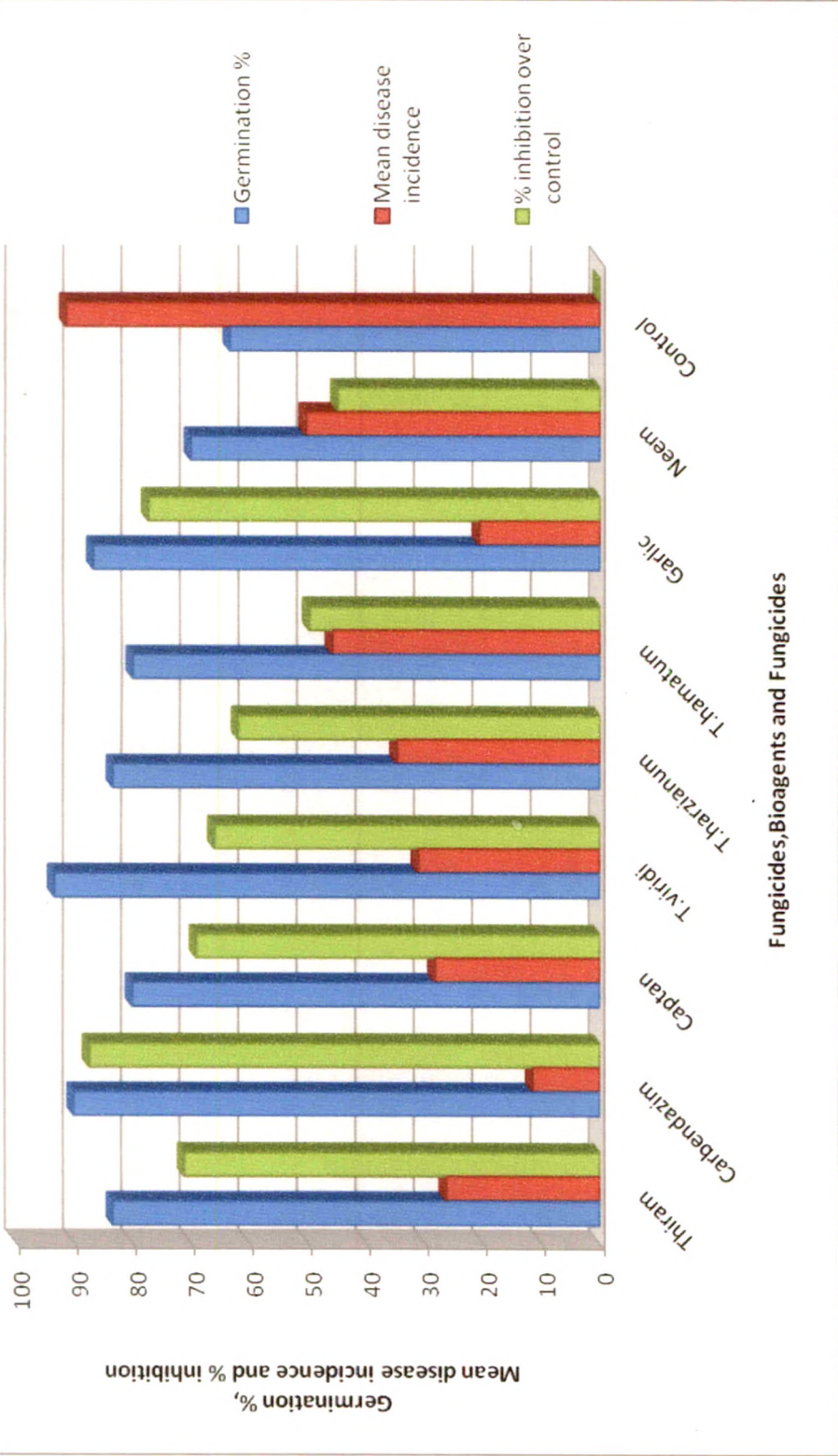
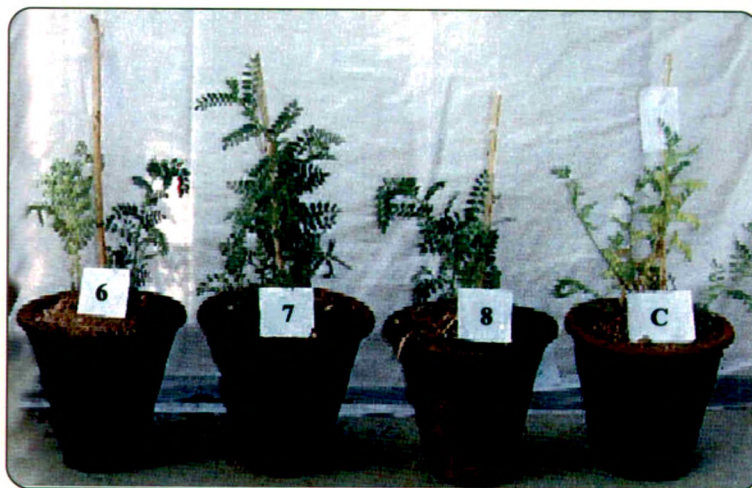
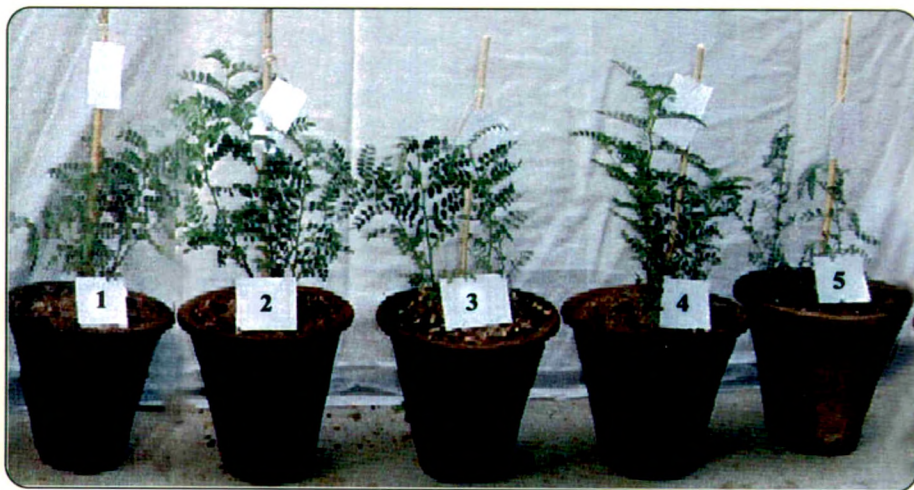


Fig 5: Effect of Seed treatment of fungicide , Bioagents and Botanicals on germination percentage of chickpea ,mean disease incidence of dry root rot of Chickpea



1. Thiram

2. Carbendazim

3. Captan

4. *T. viride*

5. *T. harzianum*

6. *T. hamatum*

7. Garlic

8. Neem

Plate 6. Effect of fungicide, botanical and bioagents seed treatment on incidence of Chickpea root rot in Pot culture

In the average dry root rot disease incidence over control, Carbendazim gave maximum disease control (87.35 per cent) which is followed by Garlic (77.27 per cent) , Thiram (71.09 per cent), Captan (69.04 per cent) , *T.viride* (65.91 per cent), *T.harzianum* (61.91 per cent), *T.hamatum* (49.75 per cent), Neem (44.91 per cent) over control (0.00 per cent).

Among three groups in Chemical group, Carbendazim was found best which gave maximum germination count (90.00 per cent) with average dry root rot incidence (11.56 per cent) and maximum disease control (87.35 per cent) as compared to thiram and Captan over control. In biocontrol group *T.viride* was found most effective as compared to other two bioagents (i.e. *T.harzianum* ,*T.hamatum*) and *T.viride* gave maximum germination count with (93.33 per cent) with disease incidence (31.15 per cent) and disease control (65.91 per cent) over control. In botanicals, Garlic was found most effective as compared to Neem. Garlic gave maximum germination count (86.67 per cent) and least disease incidence (20.77 per cent) and control (77.27 per cent).

The efficacy of fungicides, botanicals and bioagents were almost of similar under in vitro and pot culture evaluation. Present studies were similar to those observed by Chauhan *et al.*(1981), Bhattacharya *et al.* (1956), Datar (1999), Kumar *et al.* (2007).

4.8 Evaluation of Chickpea genotype against dry root rot

It is evident from the table 8 that out of 35 germplasm line, 13 lines viz AKG1106, SAKI9516, PG8108, BCP26, BDNG20B-1, ICCV0821, ICCV04608, BCP49, BCP21, JG315 (R-ch), Rajas, PG0515, BDNG 13-1 were found resistant. The seven line viz PG0302-10, Digvijay, AKG 1108, Vishal, PG 611-14, JAKI 9218, ICCV-03-407 were found moderately resistant. The

four line viz BDNG 797, BDNG 9-3, ICCV07313, ICCV-5313 were found moderately susceptible. The two line viz AKG 1103, Vijay were found susceptible. The nine lines viz BDNG 2013-2, PG9801,JG-62(S-W.Ch), PG-04305, PG0014-6-3, PG-0102-11-2, PG-00381, PG-9758, PG4333 were found highly susceptible. Differential responses of chickpea genotypes to dry root rot have been reported by Singh and Mehrotra (1982),Nene *et al.*(1989), Reddy *et al.* (1991), Bekele *et al.* (1992), Bhatt (1993), Gangwar *et al.*(2002), Pande *et al.* (2004), Katariya and Gaur (2007) and Jaylakshmi *et al.* (2008).

Table -6 Evaluation of Chickpea Genotype against root rot

Sr. No.	Genotypes	Incidence of root rot (%)	Disease Reaction
1	PG0302-10	13.00	MR
2	Digvijay	17.51	MR
3	AKG 1106	9.79	R
4	AKG 1103	49.42	S
5	AKG 1108	10.94	MR
6	SAKI 9516	7.83	R
7	PG 8108	5.80	R
8	BCP 26	4.02	R
9	Vishal	16.23	MR
10	PG 611-14	14.12	MR
11	BDNG 13-1	9.92	R
12	BDNG 2013-2	54.14	HS
13	BDNG 20B-1	6.57	R
14	JAKI 9218	13.11	MR
15	BDNG797	23.33	MS

Table 6 contd.....

Sr. No.	Genotypes	Incidence of root rot (%)	Disease Reaction
16	BDNG 9-3	21.09	MS
17	Vijay	39.24	S
18	ICCV 0821	07.63	R
19	ICCV04608	03.35	R
20	PG 9801	66.67	HS
21	ICCV07313	29.04	MS
22	BCP49	06.46	R
23	BCP 21	6.02	R
24	JG 315 (R-Ch)	6.44	R
25	JG 62 (S-W.Ch)	100	HS
26	PG-04305	53.33	HS
27	PG-0014-6-3	66.67	HS
28	PG-0102-11-2	68.89	HS
29	ICCV-5313	24.44	MS
30	PG-00381	100.00	HS
31	Rajas	6.66	R
32	PG-4333	77.78	HS
33	ICCV-03-407	16.69	MR
34	PG-9758	100	HS
35	PG-0515	10	R

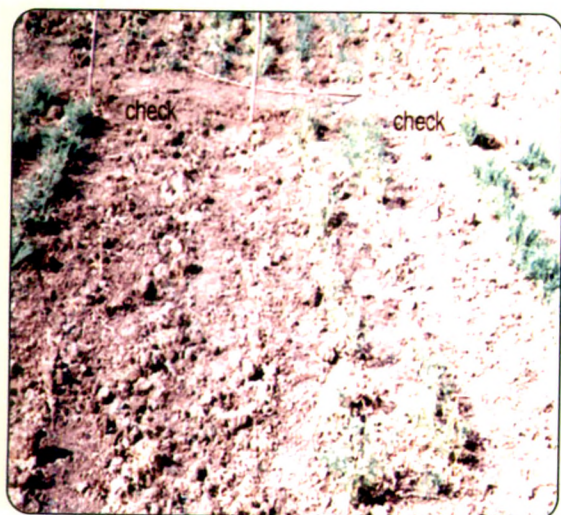
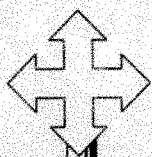
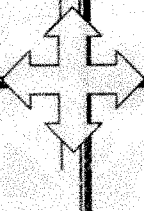


Plate 7. Field screening



SUMMARY AND CONCLUSION



CHAPTER V

SUMMARY

The cultivated chickpea (*Cicer arietinum* L) was one of the first grain legume to be domesticated in the old world. The chickpea crop is attacked by 172 pathogen, 67 fungi, 22 viruses, 3 bacteria, 80 nematodes and mycoplasma from all over the world (Nene *et al.* 1996). Among all only a few of them have the potential to devastate the crops. Some of the serious disease in order of their importance are wilt (*Fusarium oxysporum* f. sp. *ciceri*), Dry root rot (*Rhizoctonia bataticola*), Ascochyta blight (*Ascochyta rabici*) and collar rot (*Sclerotium rolfsii*). Among all dry root rot caused by *Rhizoctonia bataticola* (Taub) Butler is one of the major soil born disease of chickpea.

Typical symptoms of the disease is plants becomes straw colored as dry grass at pod formation and grain filling stage, root becomes black, rotted and brittle. Shredding of the bark, roots broken easily, Minute dark black sclerotial bodies on root surface and in the pith region.

A roving survey for recording chickpea dry root rot disease incidence was undertaken in Marathwada region. The highest disease incidence was observed in Latur district (23 per cent) and lowest in Aurangabad district (10 per cent).

The fungus (*R. bataticcola*) was isolated from naturally infected diseased roots on Potato Dextrose Agar. The fungus produced grey to black colored colony. Pathogenicity of the *R. bataticcola* was proved on one month old seedlings of Chickpea JG-62.

In *in vitro* total eight fungicides, were evaluated at the different concentration. The pathogen was most sensitive to Carbendazim + Mancozeb (100 per cent inhibition) followed by Thirum + Carbendazim (99.22 per cent inhibition), Carbendazim (97.74 per cent inhibition), Captan (90.98 per cent inhibition) and Mancozeb (90.55 per cent inhibition). Thirum recorded 78.21 per cent mycelial inhibition followed by copper oxy chloride (40.60 per cent inhibition). And least inhibition was recorded by Zineb (32.42 per cent). The clove extract of Garlic (*Allium sativum*) (75.25 per cent) produced maximum growth inhibition of pathogen.

followed by finger extracts of Turmeric(*curcuma longa* L.) (63.98 per cent) and rhizome extract of Ginger (*Zingiber officinale* L) (60.75 per cent). The next in order of merit were leaves extract of Mehandi (55.42 per cent),leaves extract of Bouganvella (48.97 per cent) ,leaves extract of Parthenium (34.44 per cent).Whereas, leaves extract of Lantana(*Lantana camera* L) (6.14 per cent) is least effective in inhibiting the growth of pathogen

The suppression of growth of pathogen was maximum with *Trichoderma viride* (86.05 per cent) followed by *Trichoderma harzianum*(58.98 per cent) and the least effective bioagents was *Pseudomonas fluoresces* (32.75 per cent).*Trichoderma viride* showed best performance against the pathogen

The three fungicides, three bioagents and two botanicals were evaluated by seed treatment method in pot culture experiment with chickpea cultivar JG 62 against *R.bataticola* .Observations on seed germination , percent disease incidence and percent inhibition over control were recorded. Among three groups in chemical group Carbendazim was found best which gave maximum germination count (90.00 per cent) with average dry root rot incidence(11.56 per cent) and maximum disease control (87.35 per cent) as compared to Thiram and Captan over control. In biocontrol group *T.viride* was found most effective as compared to other two bioagents (i.e. *T.harzianum*, *T.hamatum*) and *T.viride* gave maximum germination count with (93.33 per cent) with disease incidence (31.15 per cent) and disease control (65.91per cent) over control. In botanicals Garlic was found most effective as compared to Neem. Garlic gave maximum germination count (86.67per cent) and least disease incidence (20.77per cent) and disease control (77.27per cent).

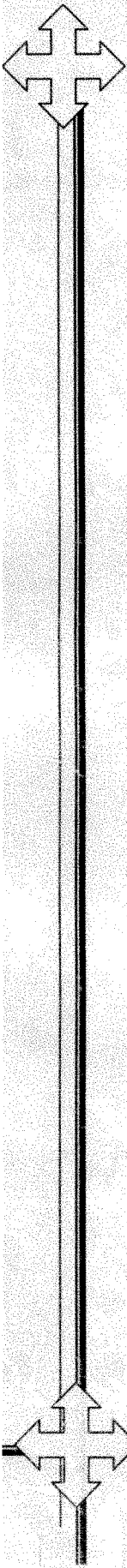
It is evident from the table 8,Out of 35 germplasm line, 13 lines viz AKG1106, SAKI9516, PG8108, BCP26, BDNG20B-1, ICCV0821, ICCV04608, BCP49, BCP21, JG315 (R-ch), Rajas, PG0515,BDNG 13-1 were found resistant. The seven line viz PG0302-10, Digvijay, AKG 1108, Vishal, PG 611-14, JAKI 9218, ICCV-03-407 were found moderately resistant. The four line viz BDNG 797, BDNG 9-3, ICCV07313, ICCV-5313 were found moderately susceptible. The two line viz AKG 1103, Vijay were found susceptible. The nine lines viz BDNG 2013-2, PG9801,JG-62(S-W.Ch), PG-

04305, PG0014-6-3, PG-0102-11-2, PG-00381, PG-9758, PG4333 were found highly susceptible.

CONCLUSIONS :

Thus from the results obtained on various aspects during present investigations on dry root rot (*R.bataticola*) of Chickpea (*Cicer arietinum*), it can be concluded that:

- *R.bataticola* is one of the most destructive pathogen causing dry root rot diseases in Chickpea and there by inflicting accountable yield losses.
- Survey report in Marathawada region showed that the dry root rot of Chickpea was severe during *Rabi* season in dry land soil.
- The pathogen (*R.bataticola*). was grow and proliferate better on a wide range of agar culture media preferably on culture media *viz.*, Potato Dextrose Agar.
- Studies (*In vitro*) on efficacy of fungicides, bioagents and plant extracts against *R.bataticola* proved that, Carbendazim+ Mancozeb , Thiram + Carbendazim, Carbendazim are effective fungicides to control the dry root rot of Chickpea. Among the bioagents *T. viride* is best to control dry root rot followed by *T. harzianum* and in plant extract Garlic extract and Turmeric bulb extract could effectively and economically manage/control dry root rot of chickpea.
- In pot culture studies, among three groups in chemical group Carbendazim was found best which gave maximum germination count (90.00 per cent) with average dry root rot incidence (11.56 per cent) In biocontrol group, *T.viride* was found most effective. In botanicals, Garlic was found most effective.
- In field screening out of 35 germplasm line 13 lines were found resistant. The seven line were found moderately resistant. The four line were found moderately susceptible. The two line were found susceptible. The nine lines were found highly susceptible.



LITERATURE CITED

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- Agrawal and Sarbhoy, A.K. 1976. Evaluation of soybean for resistance against *Rhizoctonia bataticola*. *Indian Phytopath.* 29:190-191.
- Ahmed, Q and Mohamad, A. 1986. Losses in yield due to *Rhizoctonia* root rot of Chickpea in Bihar. *Indian Phytopath.* 39:590-592
- Bensal, R.K., Sobati, A.K. and Mehta S.M. 1990. Cultural variability among seven isolates of *Rhizoctonia bataticola*. *Indian Phytopath.* 43:448-450.
- Beniwal, S.P.S., Ahmed, S. and Gorfu, D. 1992. Wilt and root rot diseases of Chickpea in Ethiopia. *Trop. Pest Management.* 38(1):48-51
- Bhate, D.K. 1983. Study of charcoal rot of Green chickpea. M.Sc (Agri.) thesis submitted to MPKV Rahuri, Dist Ahmednagar, Maharashtra.
- Bhatt, J. 1993. Reaction of chickpea cultivars to *R. bataicola* (Taub) Butler. *Indian J. Pulses. Res.* 6 (1):118-119.
- Casas, A.T. and Diaz, R.M.J. 1985. Fungal wilt and root rot diseases of chickpea in southern Spain. *Indian Phytopath.* 75(10) : 1146-1150.
- Chaturvedi, S.K. and Dua, R.P. 2009. Improved varieties of chickpea in India. *Bull, Kanpur* : 1-10.
- Chaudhary, S., Pareek, S. and Sexena, J. 2010. Efficacy of biocontrol agents singly and in combination against dry root rot. *Macrophomina phaseolina* of Mungbean. *J. Mycol. and Pl. Pathol.* 40 (1) : 141-144.
- Chauhan, M.S. 1962 . Study of intensity of dry root rot in sand culture. ISSN: 229-3701, 3 (4) Oct- Dec 2012.
- Chauhan, M.S. 1986. Systemic and non-systemic fungicides against root rot of cotton in Haryana. *Indian J. Mycol. and Pl. Pathol.* 16(2) : 226-227.

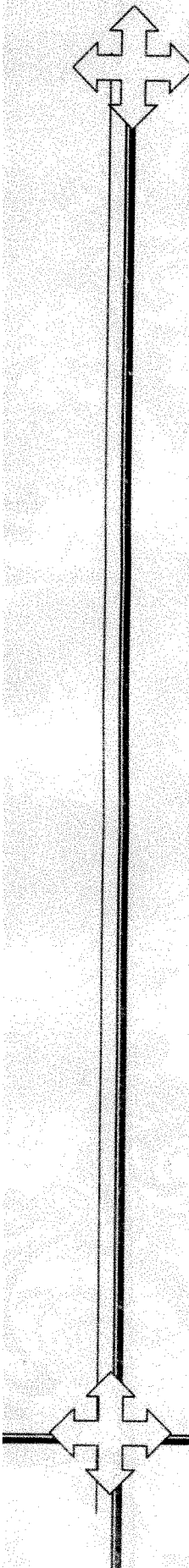
- Das, G.N. and Sen Gupta, P.K. 1963. Evaluation of fungicides in the laboratory against *Sclerotium rolfsii* Sacc. and *Rhizoctonia bataticola* (Taub) Butler. *J. Sci. and Cult.* 29(2) : 101-103.
- Datar, V.V. 1999. Bio efficacy of plant extracts against *Macrophomina Phaseolina* (Tassi) Goid. The incitant of charcoal rot of sorghum. *J.Mycol. Pl. Pathol.* 29(2): 251-253.
- Devi, P.T. and Singh, R.H. 1998. Studied on virulence of *Macrophomina phaseolina* isolates from Blackgram and Green gram. *J. Mycol. Pl. Pathol.* 28(2) : 196-201.
- Diaz, R.M.J. and Casas, T.A. 1985. Use of fungicides treatment and host resistance to control the wilt and root rot complex of chickpea. *Int. J. Trop. Pl. Dis.* 69 (7) : 591- 595.
- Dinkaran, D., Sachithanatham, K. and Nainamohammed, S. E. 2005. Screening of *Sesamum* genotypes against Dry root rot pathogen *Macrophomina phaseolina*. *Indian J. Plant Pathol.* 23 (1& 2) : 66-68.
- Dwivedi, T. S. and Chaube, H.S. 1985. Effect of fungicides on the emergence and infection on cotton seedlings by *Macrophomina phaseolina* (Tassi) Goid *Indian J.Mycol.Pl.Pathol.* 15(3):295-296.
- Gupta, I.J. and Cheema, H.S. 1990. Effect of microsclerotia of *Macrophomina haseolina* and seed dressers on germination and vigour of sesame seed. *Seed Res.* 18(2): 169-172.
- Gurha, S.N. and Trivedi Shubha. 2008. Status of soil borne pathogens infecting chickpea in Karnataka state. *Ann. Pl. Protec. Sci.* 16(1) : 255-256.
- Haware, M.P., Diaz, R.M., Amin, K.S., Phillip, J. C. and Halila, H. 1990. Integrated management of wilt and root rot of chickpea. Chickpea in nineties: Proceedings of the second International workshop on chickpea

- improvement. 8 December 1889. ICRISAT center, Patancheru, A.P., India: pp. 129-130.
- Jaylakshami, S.K.S., Usharani, V.L., Benagal and Mannur, D.M. 2008. Sources of resistance to dry root rot of Chickpea caused by *Rhizoctonia bataticola*. *Agric. Sci. Digest*. 28 (2): 147-148.
- Jyotsana Srivastava, A., Singh, G.R.P., Srivastava, A.K., Saxena, A. K. and Arora, D.K. 2008. Growth promotion and charcoal rot management in chickpea by *T. harzianum*. *J. Pl. Prot. Sci.* 48(1): 81-92.
- Katariya, L. and Gaur, V.K. 2007. Reaction of chickpea varieties for resistance to dry root rot caused by *R. bataticola*. *J. Mycol. Pl. Pathol.* 37(3):538-539.
- Kaushal, R.P. and Richa Sood .2008. Management of root rot in chickpea. *Journal of Food Legumes*. 21 (3): 178-181.
- Khan, M.A. and Gangopadhy. 2008. Efficacy of *Pseudomonas fluroescens* in controlling root rot of Chickpea caused by *Rhizoctonia bataticola*. *J. Mycol. and Pl. Pathol.* 38(3) : 580-587.
- Knock Davies. 1965. *R. bataticola* could not induce sporulation unless the culture were irradiated. ISSN : 2229-3701. 3 (4) Oct- Dec 2012.
- Kohire , O.D., Mulik, M.B., Kohire patil, V.O. and More, S.S. 2006. Survey of root disease of chickpea in Jalna Dist. of Marathawada region. *Inter. J. Pl. Protect.* 5 (2): 381-385.
- Kulkarni , S.P., Rammurthy, R. and Hegade, E. Y.1992. Growth and sclerotial production in root rot fungus of cotton. *Indian Phytopath.* 21 (4) : 79.
- Lodha, S. and Solanki, K.R. 1993. Inheritance of resistance to dry root rot in Cluster bean. *Indian Phytopath.* 45(4): 430-433.

- Malathi, P. and Doraisamy, S. 2003. Compatibility of *Trichoderma harzianum* with fungicides against *Macrophomina phaseolina*, Dis. Res. 18(2) : 139-143.
- Mishra, B.P. 1982. Evaluation of bavistin as seed treatment in controlling root rot of jute. *Indian J. Mycol. and Pl. Pathol.* 12(2):235-236.
- Nene, Y. L. 1985. Opportunities for research on diseases of pulses crops. *Indian Phytopath.* 38: 1-10.
- Pandey, G. and Singh, R.B. 1990. Survey of root rot diseases of Chickpea in Allahabad region. *Curr. Nematode.* 1(1): 77-78
- Parkhia, A.M. and Vaishnav, M.U. 1986. Biological control of *Macrophomina phaseolina*. *Indian Phytopath.* 39: 440-496.
- Prajapati, R. K. Gangwar, R. K. and Shrivastava, S.L. 2003. Resistance source of chickpea against dry root rot. *J. Farm Sci.* 12(1) : 86.
- Rana, V., Sharma, A., Paul, Y.S. and Sharma, K.D. (2009), survey for pea diseases and identification of fungi in Himachal Pradesh. *J. Mycol. and Pl. Pathol.* 39 (3) : 416-421.
- Ratan Ved and Biswas, S.K. 2010. Influence of date of sowing on incidence of dry root rot of chickpea. *Annals of Plant Protection -Science* 18 (1): 258 - 259
- Rathore, B.S. 2001. Evaluation of seed dressers against dry root rot of cluster bean. *Pl. Dis. Res.* 16 (1):89-91.
- Reddy, M.V., Raju, T.N. and Pundir, R.P.S. 1991. Evaluation of wild Cicer accessions for resistance to wilt and root rots. *Indian Phytopath.* 44 (3); 388-391. ###
- Rhiaz, A .K. ,Towseef A. B., Krishan, K. 2011. Conducted survey in Jammu and Kashmir. ISSN : 2229-3701, 3 (4) Oct-Dec 2012.
- Sajeena, A., Salaranjan, F. and Mohanbaba, R. 2004. Screening of Blackgram varieties against root rot caused by *Macrophomina phaseolina*. *J. Mycol. And Pl. Pathol.* 34: 823-825.

- Saju, K. A., Venugopal, M.N. and Mathew, M.J. 1998. Antifungal and insect repellent. 7 : 660-602.
- Saxena, D.R. and Saxena, M.1993. Evaluation of chemicals against biological drying of lentil. *Bhartiya krishi anusandhan Patrika*. 8 (1) : 25-29.
- Sharma, B. L., Gupta, R.N. and Gupta, J.S. 1983. Studies on survey of wilt and root rot incidence of *Cicer arietinum* in northern region of Madhya Pradesh. *Indian Phytopath.* 36(1): 82-84.
- Sharma, O.P. and Gupta, R.B.L. 2004. Fungicides in the control of chickpea dry root rot caused by *R. bataticola* *J. Mycol. and Pl. Pathol.* 34(2) : 321-322.
- Shukla, B.N. and Singh, B.P. 1973. Effect of fungicidal seed treatments on *Macrophomina* root rot of sesamum. *Indian Phytopath.* 3(2) : 208.
- Singh, A. and Sirohi A. 2003. Status of chickpea diseases in Himachal Pradesh, India. *ICPN*. 10 : 29-31.
- Singh, P.J., Yadav, J.P. and Mehrotra, R.S. 1982. Commission of mycelial inoculums of *Rhizoctonia bataticola* (*Macrophomina phaseolina*) Qeecting damping off & root rot of chickpea. *Indian Phytopath.* 4(2) : 104.
- Singh, R. and Sindhan, G.S. 1998. Effect of fungicides on the incidence of dry root and biochemical status of Chickpea plants. *Indian J .Mycol. And Pl. Pathol.* 13(1) : 14-17.
- Sobati, A.K. and Bansal R. K. 1988. Cultural variability among three isolates of *R. bataticola* from Groundnut. *Indian Phytopathol.* 41:149-151.
- Taya, R.S., Tripathi, N.N. and Panwar, M.S.1988. Influence of texture and nutritional status of soil on the control of dry root rot of chickpea. *Indian J. Mycol. And Pl. Pathol.* 20(1): 14-20.

- Vijay Mohan, Prasad , S.M., Barnwal, M. K. and Kudada, N. 2006 Fungicidal management of dry root rot disease and yield of chickpea. *J. Appl. Biol.* 16(1-2) : 42-44.
- Vishwadhar and Sarbhoy, S.K. 1991. An typical isolates of *Rhizoctonia bataticola*. *Indian Phytopath.* 26:245-246.
- Zote *et al.* 1993. Source of resistance to Fusarium wilt and dry root rot of Chickpea. *International Chickpea Newsletter.* (29): 11-12.



THESIS ABSTRACT

ABSTRACT

“Investigation on Dry root rot of Chickpea incited by *Rhizoctonia bataticola* (Taub) Butler.”

By

DATE YOGINI BHAUSAHEB

A Candidate for the degree of

MASTER OF SCIENCE

In

PLANT PATHOLOGY

College of Agriculture, Badnapur

Vasantrao Naik Marathwada Krishi Vidyapeeth

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2015

Research Guide : Dr. D.N. Dhutraj

Department : Plant Pathology

The present investigation entitled “Investigation on dry root rot of chickpea (Taub) Butler.” was carried out at Plant pathology lab, College of Agriculture, Badnapur. And farm of Agriculture research station, Badnapur during the year 2014-2015. Chickpea (*cicer arietinum* L.) is one of the most important pulse crop in India. In India area under chickpea was 9.51 million ha. and production 8.88 Million ton and productivity 929 kg/ha during 2013-14 (Anonymous, 2014). Occurrence of Dry root rot in chickpea has become a major constraint for cultivation of chickpea. The efficacy of various botanicals, bioagents and fungicides was evaluated against *Rhizoctonia bataticola* (Taub) Butler. The phytoextracts and fungicides were evaluated *in vitro* by poisoned food technique against *R. bataticola*.

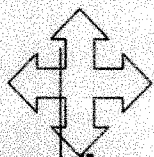
The eight fungicides were tested against *R. bataticola* by poisoned food technique *in vitro*. Carbendazim + Mancozeb, Thirum + Mancozeb were considered to be the most effective to inhibit the mycelial

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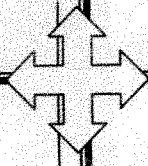
growth of pathogen as well as reducing root rot incidence, followed by Mancozeb, Thiram while Copper oxychloride and Zineb were least effective treatment. Among the tested biocontrol agents against *Rhizoctonia bataticola*, *T.viride* was found the most effective against the fungus under *in vitro* followed by *T.harzianum*, *T.hamatum*, *T. longibrachiatum*, *A.niger*, *A.flavus* while *P. fleurescens* were the least effective in reducing root rot incidence. Among the tested Botanicals against *Rhizoctonia bataticola*, Garlic found most effective followed by Turmeric while Lantana camera found least effective in inhibiting the growth of pathogen.

In pot culture, maximum percent germination was recorded in seed treatment of Carbendazim followed by Garlic, Thiram, captan *T.viride*, *T.harzianum*, *T. hamatum* while least inhibition observed in Neem.

In field screening, Out of 35 germplasm line, 13 lines viz AKG1106, SAKI9516, PG8108, BCP26, BDNG20B-1, ICCV0821, ICCV04608, BCP49, BCP21, JG315 (R-ch), Rajas, PG0515, BDNG 13-1 were found resistant. The seven line viz PG0302-10, Digvijay, AKG 1108, Vishal, PG 611-14, JAKI 9218, ICCV-03-407 were found moderately resistant. The four line viz BDNG 797, BDNG 9-3, ICCV07313, ICCV-5313 were found moderately susceptible. The two line viz AKG 1103, Vijay were found susceptible. The nine lines viz BDNG 2013-2, PG9801, JG-62(S-W.Ch), PG-04305, PG0014-6-3, PG-0102-11-2, PG-00381, PG-9758, PG4333 were found highly susceptible.



VITA



VITA

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MASTER OF SCIENCE

In

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2015

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